Development of a Biomedical Equipment Maintenance Management Software for Sri Lankan Hospitals.

by

Bumith Charuka Nallaperuma

-E217844-

Research Proposal Submitted in accordance with the requirements for the APPLIED SCIENCE RESEARCH PROJECT MODULE OF PEARSON'S HND IN BIOMEDICAL ENGINEERING PROGRAMME

at the

ESU Colombo

Name of research Tutor:-

MR. ETHIRAJ THIPAKARAN

DECLARATION

Name of Research Candidate:- Bumith Charuka Nallaperuma

Pearson Registration Number:- E217844

Programme Name:- HND Biomedical Engineering

Research Title:- Development of a Biomedical Equipment Maintenance Management Software for Sri Lankan Hospitals

Field of Study:

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Supervisor's Name: Designation:

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Thank you, sir, for your support and for believing in me

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E217844.

ABSTRACT

Effective maintenance and fault reporting of biomedical equipment is essential to ensuring optimal healthcare service delivery, safety, and operational efficiency within hospital environments. This project presents the development of a Biomedical Equipment Maintenance Management System (BEMMS), a web-based application designed to streamline equipment tracking, preventive maintenance scheduling, and real-time issue reporting in Sri Lankan hospitals.

The system was built using a lightweight technology stack, integrating Python with Flask for the backend, HTML/CSS and Bootstrap for the user interface, and SQLite for local database management. BEMMS enables hospital staff to log equipment issues, categorize maintenance types, and track the resolution process with automated timestamps and priority indicators. The platform provides administrators with functionalities to add, edit, and monitor both maintenance records and fault reports, significantly reducing manual documentation and the risk of equipment failure due to missed servicing.

A mixed-methods approach was adopted in the system's design and evaluation, including interviews with biomedical engineers, real-world hospital workflow analysis, and usability testing. Results indicate the system's strong potential to improve equipment uptime, enhance reporting accuracy, and support maintenance teams in proactive decision-making. This project contributes a scalable, user-friendly tool that aligns with biomedical equipment management best practices and serves as a foundation for future integration with QR code-based asset tracking and automated alert systems.

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CHAPTER 1 – INTRODUCTION

1.1 Introduction

The efficient maintenance of biomedical equipment is a fundamental requirement for delivering safe, reliable, and timely healthcare services. In hospital environments, where life-supporting and diagnostic machines such as defibrillators, infusion pumps, ventilators, and dialysis systems are integral to clinical outcomes, equipment downtime can directly compromise patient care and safety. As healthcare systems become increasingly reliant on complex biomedical devices, the need for structured and efficient maintenance practices grows proportionally.

In Sri Lanka, most public and semi-government hospitals continue to depend on paper-based records, logbooks, and manual reminders to manage the maintenance of medical equipment. These traditional methods are prone to delays, miscommunication, and loss of critical information, resulting in increased equipment failure rates, missed preventive maintenance schedules, and reduced operational readiness during emergencies. Biomedical engineers, who are responsible for the maintenance, calibration, and fault diagnosis of equipment, often lack access to modern digital tools that could support their decision-making and streamline workflow.

In contrast, many international healthcare systems have already adopted Computerised Maintenance Management Systems (CMMS) to improve asset tracking, maintenance scheduling, inventory management, and compliance reporting. However, existing commercial CMMS solutions are typically expensive, complex, and not tailored to the infrastructure limitations or staffing capacities of Sri Lankan hospitals. This creates a critical technological gap in the country's healthcare sector, particularly affecting equipment reliability, maintenance visibility, and knowledge transfer among biomedical engineering teams.

A real example can be drawn from Hospital, where biomedical maintenance logs are still primarily handled through manual documentation. In 2023, a series of equipment failures—including a delay in the calibration of anaesthesia machines and the late servicing of infusion pumps—resulted in extended equipment downtime and disrupted surgical schedules. These delays were traced back to missed reminders and a lack of centralized maintenance tracking. Biomedical engineers at the hospital expressed the need for a more structured system that could help them respond quickly to faults and maintain accurate service records without depending solely on memory or handwritten notes.

To address such issues, the present research aims to develop a locally adaptable **Biomedical Equipment Maintenance Management Software (BEMMS)** using open-source technologies. The proposed system will integrate key functions such as QR code-based fault reporting, automated maintenance reminders, centralized equipment history logs, and support for technician training. Developed using the Django web framework with a PostgreSQL backend and deployed via Docker, BEMMS is designed to operate within the resource constraints of Sri Lankan hospitals while significantly improving maintenance efficiency and data accessibility.

This research follows a pragmatic, mixed-methods approach, combining qualitative insights from biomedical engineers with quantitative analysis of maintenance task completion and downtime data. A pilot implementation will be conducted at selected hospital sites to evaluate the software's usability, real-world applicability, and measurable benefits. The overarching objective is to provide a scalable, cost-effective digital tool that enhances the operational effectiveness of biomedical engineers and contributes to the modernization of Sri Lanka's healthcare infrastructure.

1.2 Purpose of the Research

The primary purpose of this research is to design, develop, and evaluate a tailored Biomedical Equipment Maintenance Management Software (BEMMS) to enhance the efficiency, traceability, and responsiveness of medical equipment maintenance in Sri Lankan hospitals. This study addresses a well-documented gap in the country's healthcare infrastructure, where maintenance processes are still predominantly manual and fragmented, posing serious risks to equipment reliability, patient safety, and operational continuity.

The proposed software aims to introduce a practical digital solution that automates and centralizes the maintenance workflow of biomedical equipment. Specifically, the system is intended to support biomedical engineers and technical staff in public and semi-government hospitals by offering the following key functionalities:

- QR code-based fault reporting, enabling hospital staff to immediately report equipment issues with traceable identifiers;

- Automated maintenance scheduling and alerts, reducing the risk of missed preventive service intervals;
- Equipment service history tracking, providing detailed digital logs accessible to both senior and newly appointed biomedical engineers;
- Image upload functionality, supporting remote diagnostics and detailed fault documentation.

The software will be implemented and evaluated in a real-world hospital setting, beginning with a pilot at the Hospital, to assess its impact on key performance indicators such as equipment downtime, response time to breakdowns, and user satisfaction.

By integrating modern web technologies such as Python, Django, and PostgreSQL with hospital-specific workflows, the project seeks not only to solve an immediate operational challenge but also to create a sustainable, replicable solution that could be scaled across the national healthcare system. The long-term vision is to contribute to the digital transformation of biomedical engineering practices in Sri Lanka, offering improved safety, accountability, and training support within healthcare institutions.

This purpose is directly aligned with the broader theme of this unit, delivering scientific projects in the real world, and exemplifies how applied science and technology can be harnessed to solve practical challenges in healthcare environments.

1.3 Significance of the Research

This research carries significant value for the Sri Lankan healthcare system, particularly in enhancing the quality, reliability, and efficiency of biomedical equipment maintenance in public and semi-government hospitals. Medical devices are central to diagnostics, treatment, and patient monitoring. Therefore, any delays in maintenance or fault rectification directly impact clinical outcomes and patient safety. Despite this, many hospitals in Sri Lanka lack structured digital maintenance systems, leading to critical equipment failures, long repair lead times, and uncoordinated recordkeeping.

The development of the Biomedical Equipment Maintenance Management Software (BEMMS) represents a timely and innovative solution to these problems. By replacing outdated manual methods with a modern, open-source software system tailored to local hospital workflows, the project aims to significantly improve the management of preventive and corrective maintenance activities.

The key areas where this research is expected to make a measurable impact include:

- Operational efficiency - Automating maintenance reminders and fault reporting ensures that critical equipment receives timely attention, minimising unplanned downtime and improving hospital readiness.
- Data-driven decision-making - Access to historical equipment maintenance data allows biomedical engineers and administrators to identify patterns, prioritise replacements, and allocate resources more effectively.
- Training and skill development - The system's ability to log detailed repair histories serves as an educational tool for newly appointed engineers and technicians, helping bridge the skill gap in the biomedical engineering workforce.
- Cost-effectiveness and scalability - Unlike imported CMMS tools that are often too costly or complex for local adoption, BEMMS is designed using free, open-source technologies, ensuring affordability, adaptability, and replicability across hospitals in the country.
- Policy and standardization - With successful pilot implementation and adoption, the findings of this research could inform national-level strategies for biomedical equipment maintenance, setting a precedent for standardized digital practices in healthcare infrastructure.

This study is particularly significant in the post-pandemic era, where the resilience and reliability of healthcare systems have become critical priorities. By addressing a deeply rooted operational weakness in Sri Lanka's public health sector, this research stands to contribute meaningfully to long-term improvements in healthcare delivery, workforce efficiency, and patient safety.

1.4 Research Objectives

The overarching aim of this research is to develop and evaluate a context-specific Biomedical Equipment Maintenance Management Software (BEMMS) to enhance maintenance practices in

Sri Lankan hospitals. The system is intended to address the limitations of manual tracking methods by providing a digital, automated, and user-friendly platform for biomedical engineers and hospital staff.

The study is guided by the following four main research objectives

RO1: To analyze the current practices and key challenges associated with biomedical equipment maintenance in Sri Lankan hospitals, with a particular focus on scheduling inefficiencies, communication gaps, and documentation limitations.

RO2: To design and propose a practical, scalable, and locally adaptable software model for biomedical equipment maintenance management, incorporating user-focused features such as QR code-based fault reporting, automated alerts, and centralized service history tracking.

RO3: To assess the effectiveness of the BEMMS system by comparing pre- and post-implementation data, measuring improvements in equipment uptime, fault response time, preventive maintenance compliance, and overall user satisfaction.

RO4: To evaluate the anticipated benefits of implementing the proposed software in terms of improved maintenance efficiency, increased equipment uptime, and reduced operational costs, based on measurable outcomes during the pilot phase.

These objectives form the structural backbone of the research and will be addressed through a combination of qualitative and quantitative data collection methods under a pragmatic mixed-methods framework.

1.5 Research Sub-Objectives

To support the achievement of the main research objectives, the following sub-objectives have been identified. These sub-objectives break down each research aim into actionable components that will guide data collection, system design, implementation, and analysis throughout the project:

SO1: To document and evaluate existing manual maintenance procedures, including the use of logbooks, spreadsheets, and informal reporting channels.

SO2: To develop a working prototype of the BEMMS platform using Django, PostgreSQL, and QR code integration, tailored to the workflow of Sri Lankan hospital environments.

SO3: To collect and analyze quantitative and qualitative data on system usability, user satisfaction, and integration into routine maintenance processes.

SO4: To estimate the potential long-term cost savings and operational benefits achievable through continued use of the BEMMS system, with a focus on scalability across public hospitals.

1.6 Research Questions

The research questions serve as the foundation for guiding the inquiry, data collection, and evaluation process of this study. Each question is aligned with a corresponding research objective to ensure a systematic approach to achieving the goals of the project.

1. What are the existing practices and key challenges faced by biomedical engineers in managing medical equipment maintenance within Sri Lankan hospitals, particularly regarding scheduling, communication, and documentation?

2. What essential features and functionalities should a locally adaptable biomedical equipment maintenance software include to effectively support the operational needs of hospital biomedical engineering teams?

3. How effective is the proposed BEMMS system in improving preventive maintenance scheduling, fault reporting, service tracking, and overall user satisfaction when implemented in a real-world hospital environment?

4. What measurable improvements, such as increased equipment uptime, faster response times, and reduced operational costs, can be observed after implementing the BEMMS system during the pilot phase?

1.7 Hypothesis

The development and implementation of a Biomedical Equipment Maintenance Management Software (BEMMS) is expected to significantly improve the efficiency, reliability, and responsiveness of medical equipment maintenance in Sri Lankan hospitals. To formally investigate the anticipated impact of the system, the following hypotheses are proposed:

H_1 (Alternative Hypothesis)

The implementation of a locally adapted biomedical equipment maintenance software (BEMMS) will significantly improve maintenance efficiency, increase equipment uptime, enhance fault reporting response time, and reduce operational costs in Sri Lankan hospitals.

H_0 (Null Hypothesis)

The implementation of the BEMMS software will not result in any significant improvement in maintenance efficiency, equipment uptime, fault response time, or operational cost reduction in Sri Lankan hospitals.

1.8 Thesis Structure

This research report is structured into five main chapters, each addressing a specific aspect of the study, from problem identification to implementation and evaluation of the proposed Biomedical Equipment Maintenance Management Software (BEMMS). The chapter-wise structure is as follows:

Chapter 1
Introduction

Provides the background to the research problem, outlines the research aim, objectives, sub-objectives, research questions, hypothesis, and presents the overall structure of the thesis. It introduces the real-world significance of improving biomedical equipment maintenance through digital tools in Sri Lankan hospitals.

Chapter 2
Literature Review

Critically examines existing literature on biomedical equipment maintenance practices, Computerised Maintenance Management Systems (CMMS), and technology adoption in low-resource healthcare settings. It also highlights the research gap, conceptual framework, and rationale for the proposed software solution.

Chapter 3
Methodology

Describes the research philosophy, approach, strategy, and the mixed-methods design used in the study. It details data collection procedures, sampling strategy, tools for analysis, system development methods, ethical considerations, and how reliability and validity were ensured throughout the research.

Chapter 4
Results and Discussion

Presents the findings obtained from the pilot implementation of BEMMS in a selected hospital setting. This includes demographic data, pre- and post-intervention performance comparisons, correlation and regression analyses, and a discussion of how the outcomes address each research objective.

Chapter 5
Conclusions and Recommendations

Summarizes the key findings in relation to the research objectives and questions, presents practical recommendations for hospital administrators and policymakers, outlines limitations of the study, suggests future improvements, and includes a reflective commentary on the research journey and its contribution to industry

Chapter 2 – Literature Review

2.1 Literature Review

Introduction

Biomedical equipment forms the technological backbone of modern healthcare systems. From diagnostic imaging machines to vital sign monitors, the effective functioning of this equipment is crucial for ensuring patient safety, enhancing treatment outcomes, and supporting clinical workflows. However, the performance and reliability of such equipment depend heavily on systematic and timely maintenance practices. In many low- and middle-income countries (LMICs), including Sri Lanka, biomedical equipment maintenance is still predominantly managed through manual records and ad hoc procedures. This literature review critically examines global practices in biomedical equipment maintenance, evaluates the limitations encountered in Sri Lankan hospitals, and justifies the need for a locally adapted software solution to address identified gaps.

Global Practices in Biomedical Equipment Maintenance

In developed healthcare systems, Computerised Maintenance Management Systems (CMMS) have revolutionized the way medical equipment is tracked, scheduled, and serviced. These systems automate preventive maintenance cycles, enable real-time fault logging, maintain complete service histories, and even integrate with inventory and procurement systems. Hospitals implementing CMMS platforms report significant improvements in equipment uptime, regulatory compliance, technician efficiency, and patient safety (Gupta et al., 2022).

Several well-established platforms, such as IBM Maximo, Infor EAM, and Accruent's Maintenance Connection, are widely adopted across North America and Europe. These tools are often complemented with Internet of Things (IoT) integrations that allow for real-time condition monitoring and predictive maintenance strategies. According to Mehta and Kumar (2020), predictive CMMS implementation has been associated with a 30–50% reduction in equipment failures and notable improvements in technician response times.

Furthermore, the World Health Organization (WHO, 2010) strongly advocates for the digitization of biomedical asset management, particularly in public health institutions. The WHO has noted that over 50% of biomedical equipment in developing nations is either underutilized or non-functional due to insufficient maintenance systems. Proper implementation of CMMS is shown to improve traceability, accountability, and budget utilization through lifecycle data analytics and optimized scheduling.

Limitations of CMMS in Low-Resource Settings

Despite their advantages, CMMS platforms designed for high-income countries often present challenges when deployed in LMICs like Sri Lanka. One major barrier is the cost of acquisition and upkeep. Licensing fees, mandatory maintenance contracts, and server requirements pose significant hurdles for underfunded public hospitals. Additionally, many CMMS platforms assume access to uninterrupted internet, dedicated IT support, and skilled technical personnel—all of which are limited in Sri Lanka's state-run hospitals (Jayawardena, 2017).

Usability also emerges as a key constraint. Many of these systems are designed with complex dashboards and technical terminologies that are not intuitive for local users. Biomedical engineers and technicians in Sri Lanka often lack formal IT training and may find international software platforms cumbersome or irrelevant to their workflows. As a result, pilot implementations of foreign CMMS solutions in local hospitals have faced resistance, abandonment, or suboptimal use.

Another structural challenge lies in the need for reliable internet connectivity and cloud infrastructure. While cloud-based CMMS solutions offer real-time synchronization and remote access, many Sri Lankan hospitals—particularly those outside urban centers—face internet instability or restricted bandwidth. These technical limitations hinder smooth adoption and necessitate offline-compatible, lightweight systems.

Biomedical Equipment Maintenance in Sri Lanka

In Sri Lanka, biomedical equipment maintenance practices remain predominantly manual and decentralized. Most government hospitals rely on handwritten logbooks, Excel spreadsheets, and verbal communication to manage fault reports and preventive maintenance schedules (Malalasekara, 2024). This results in frequent service delays, poor record-keeping, and a lack of visibility into equipment histories.

A multi-hospital assessment conducted by Chaminda et al. (2023) revealed a widespread absence of standardized maintenance protocols. Biomedical departments often struggle to execute timely preventive maintenance due to inconsistent scheduling, limited technician availability, and bureaucratic procurement delays. Many hospitals also lack centralized asset registries, forcing engineers to rely on personal records or memory when tracking service histories and equipment locations.

Edirisingha et al. (2023) noted that the absence of digital tracking systems leads to duplicate work orders, unmonitored faults, and increased risk of equipment-related adverse events. The lack of real-time communication between clinical staff and maintenance teams further exacerbates delays, as fault reporting is often delayed or incomplete.

Additionally, spare parts inventory is generally managed manually. Parts availability is not digitally recorded, and restocking is reactive rather than predictive, leading to long repair lead times. These inefficiencies negatively impact clinical services, particularly in critical care areas such as operating theatres, emergency rooms, and intensive care units.

Regional Best Practices and Lessons from LMICs

While Sri Lanka continues to face systemic maintenance challenges, other LMICs have made notable progress through locally tailored CMMS solutions. A strong example is India's *e-Upkaran* platform, developed by the National Health Systems Resource Centre (NHSRC) and deployed across public hospitals in Rajasthan. The platform includes modules for complaint registration, technician assignment, preventive maintenance reminders, and equipment utilization tracking.

According to NHSRC (2021), e-Upkaran helped reduce downtime of critical medical equipment by over 40% within two years of implementation. The system's success is largely attributed to its low hardware requirements, user-friendly interface, and consultative design process, which involved hospital engineers and clinical users during development.

These regional success stories suggest that contextual design and local adaptability are key to CMMS effectiveness in resource-limited settings. Simply replicating systems from high-income countries is often counterproductive. Instead, custom-built solutions that reflect ground-level workflows, infrastructure limitations, and user capabilities offer a more sustainable path forward.

Proposed Solution: BEMMS

In response to these localized challenges, this study introduces the Biomedical Equipment Maintenance Management Software (BEMMS)—a lightweight, modular software system developed using open-source technologies such as Python Django and PostgreSQL. The system is designed specifically for Sri Lankan hospitals, with key features informed by extensive stakeholder feedback from biomedical engineers, technicians, and hospital administrators.

Key modules of BEMMS include

- QR code-based fault reporting - Hospital staff can scan QR codes affixed to equipment and instantly submit fault reports with optional images.

- Automated preventive maintenance scheduling - Using task schedulers like django-crontab, BEMMS alerts technicians in advance for each scheduled service.
- Centralized service history - Equipment logs are stored digitally, allowing traceability of past repairs, parts used, and technician assignments.
- Role-based access - Interfaces are customized based on user roles, administrators, engineers, or department heads.
- Offline functionality - BEMMS includes fallback modes for low-connectivity environments.

Theoretical and Methodological Underpinnings

The BEMMS initiative is informed by the Socio-Technical Systems Theory, which emphasizes the mutual adaptation between technical tools and social environments. This perspective highlights the need to align software functionality with human capabilities, organizational culture, and contextual limitations. In Sri Lanka's hospital context, where administrative bureaucracy, limited IT training, and constrained resources prevail, a socio-technical approach is vital.

In parallel, the Action Design Research (ADR) methodology has guided the iterative development and testing of BEMMS. ADR blends the cycles of system development with continuous real-world feedback, allowing software refinements in parallel with pilot implementation. This ensures the final product is both technically functional and socially acceptable.

Explaining the Research Gap

Despite the global recognition of Computerised Maintenance Management Systems (CMMS) as essential tools for ensuring the reliability, traceability, and efficiency of biomedical equipment maintenance, there remains a significant disparity between their availability and effective implementation in low- and middle-income countries (LMICs) such as Sri Lanka (Gupta et al., 2022; WHO, 2010). A critical analysis of existing literature reveals that most studies and commercially available CMMS tools are developed with high-income healthcare environments in mind, often assuming the presence of robust infrastructure, continuous internet access, and highly trained personnel (Jayawardena, 2017).

These assumptions do not align with the realities of Sri Lankan public hospitals, where maintenance is still largely managed through handwritten logs, verbal communication, and basic spreadsheets (Malalasekara, 2024). This results in frequent delays, lack of service history, and missed preventive maintenance tasks (Chaminda et al., 2023). Despite the recognition of these limitations, no research has yet proposed or piloted a scalable, low-cost, context-specific software solution developed with and for Sri Lankan biomedical professionals.

Furthermore, existing systems generally fail to engage end-users—particularly biomedical engineers and technicians—in the co-design process. As a result, imported CMMS platforms often suffer from usability issues and resistance to adoption due to their poor alignment with the actual workflows of these users (Edirisingha et al., 2023). These concerns are widely acknowledged but remain largely unaddressed in the academic and technical literature.

Thus, the research gap lies in the absence of locally designed, user-focused biomedical maintenance software for Sri Lankan hospitals, particularly one developed using open-source tools and validated through field implementation. This study addresses that gap by proposing and evaluating a Biomedical Equipment Maintenance Management Software (BEMMS) that reflects the real needs of Sri Lankan hospital environments and frontline users.

The literature clearly shows that while CMMS platforms have transformed maintenance operations in developed countries, these solutions are not directly transferable to the Sri Lankan context. The unique constraints faced by Sri Lankan public hospitals, including budget limitations, skill gaps, and weak infrastructure, require software solutions built from the ground up with local conditions in mind.

BEMMS emerges as a promising response, combining simplicity, affordability, and critical maintenance features that are aligned with real hospital workflows. Through pilot testing and feedback analysis, this study aims to validate BEMMS as a viable model for broader adoption and to contribute new insights to the field of biomedical engineering management in LMICs.

2.2 Conceptual Framework

The conceptual framework serves as the structural foundation guiding the design, development, and evaluation of the Biomedical Equipment Maintenance Management Software (BEMMS) in the context of Sri Lankan public hospitals. It outlines the relationship between existing biomedical maintenance challenges and the proposed digital intervention, using theoretical insights and practical workflows to support research implementation.

Theoretical Foundations

This research draws upon two well-established theoretical models:

- **Socio-Technical Systems Theory**: Emphasizes the need for a balanced integration of social and technical subsystems in healthcare environments. BEMMS is developed to fit the workflows, skills, and infrastructure found in Sri Lankan hospitals.
- **Action Design Research (ADR)**: Supports the iterative development and real-time testing of digital solutions in practice. Through ADR, BEMMS is designed with user feedback loops and continuous improvement cycles during its pilot deployment.

Conceptual Framework Table

Component	Description	BEMMS Role
Biomedical Equipment Assets	Includes all clinical diagnostic and treatment devices requiring maintenance.	Tracks each asset's history, faults, and scheduled servicing.
Hospital Environment	Reflects budget limitations, poor internet, and limited IT support common in public hospitals.	Built using open-source tools with offline functionality and low resource use.
Human Actors	Biomedical engineers, technicians, administrators, and clinical staff.	Role-based access system for tailored interfaces and task assignments.
Maintenance Process Challenges	Manual records, missed preventive tasks, and unstructured reporting.	Automates scheduling, enables QR fault reporting, and centralizes data.

BEMMS Digital Intervention	The core software system designed to improve equipment uptime and fault management.	Integrates modules that address all above pain points in one platform.
Expected Outcomes	Increase in uptime, improved preventive compliance, faster response times, better documentation.	Measured during pilot phase and analyzed in post-implementation evaluation.

Table 1 Conceptual Framework Table

Logical Flow of the Framework

The BEMMS framework begins by identifying the inefficiencies in current biomedical maintenance operations. It then introduces BEMMS as a targeted intervention that integrates with the local context, users, and processes. The resulting outputs such as higher uptime, reduced delays, and improved documentation form the basis for evaluating its success

Link to Research Objectives

This conceptual structure supports each of the main objectives:

- **RO1**: Analyzes current challenges in manual maintenance systems.
- **RO2**: Guides the design of a software solution suited to local infrastructure.
- **RO3**: Enables performance measurement through pilot implementation metrics.
- **RO4**: Provides the basis to evaluate cost efficiency, uptime, and maintenance effectiveness.

This framework enables the development and validation of a context-specific software tool that directly addresses the unique challenges of Sri Lankan hospitals. Grounded in theory and adapted through real-world feedback, BEMMS offers a promising, evidence-based solution to bridge the digital gap in biomedical equipment maintenance management.

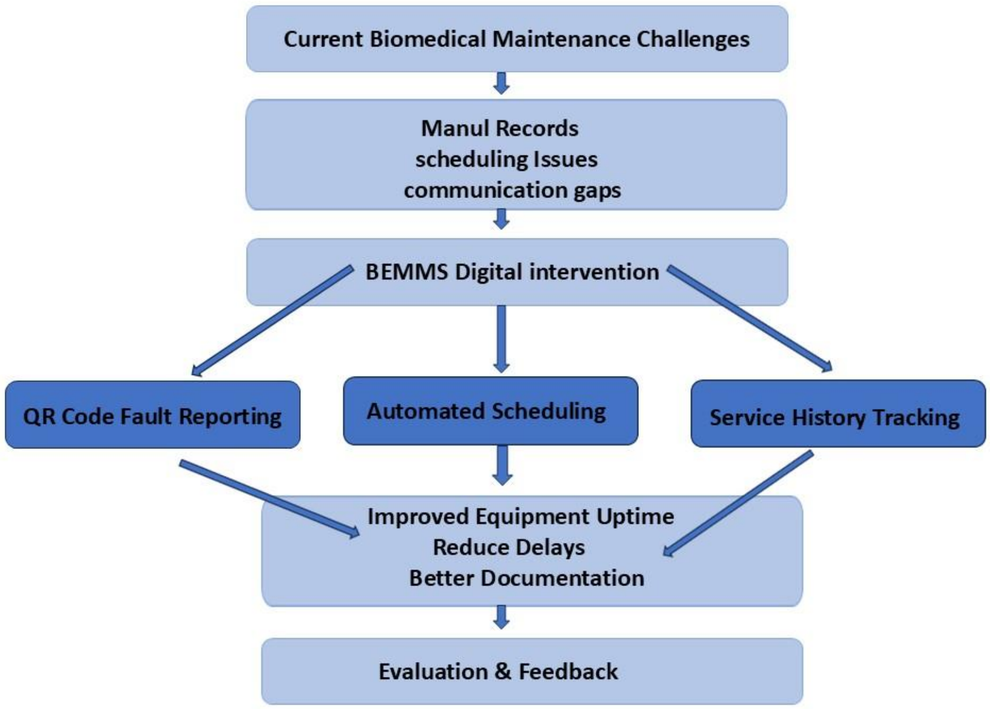


Figure 1 Conceptual Framework Flowchart

CHAPTER 3 – METHODOLOGY

3.1. Research philosophy

The research philosophy selected for this study is Pragmatism, which emphasises the practical application of knowledge and accepts that reality can be interpreted in multiple ways depending on the research context. Pragmatism supports the integration of both qualitative and quantitative data to solve real-world problems effectively making it ideally suited for applied projects like the development and evaluation of the Biomedical Equipment Maintenance Management Software (BEMMS).

Pragmatism does not align exclusively with any single epistemological stance (positivism or interpretivism), but rather advocates for the use of methods that are best suited to answer the research questions and meet the objectives. In the context of this study, the pragmatic philosophy allowed the researcher to:

- Understand real-life problems faced by biomedical staff in Sri Lankan hospitals through qualitative interviews and observations;
- Develop a practical solution (BEMMS) tailored to these needs using current technologies like Django and PostgreSQL
- Evaluate effectiveness through both user perceptions and measurable operational improvements (equipment uptime, maintenance records, fault response times).

By following a pragmatic approach, the study remained flexible, responsive, and outcome-focused, ensuring that the final solution was not only technically functional but also contextually appropriate and user-friendly. This aligns with the study's overall goal of producing a scalable and adaptable biomedical equipment maintenance management system for the Sri Lankan healthcare context.

3.2 Research Approach

The research approach adopted for this study is Abductive. Abduction allows for a flexible, iterative process in which observations lead to the generation or refinement of solutions, making it highly suitable for technology development and real-world problem solving.

The study began with the identification of a practical problem—the inefficiencies and limitations of biomedical equipment maintenance practices in Sri Lankan hospitals. Observations from real hospital settings, supported by literature, informed the initial assumptions about what kind of solution would be appropriate. A prototype of the Biomedical Equipment Maintenance Management Software (BEMMS) was then developed using open-source tools such as Django and PostgreSQL, incorporating features like QR-code fault reporting and automated preventive maintenance scheduling.

Once the system was developed, it was tested and refined through pilot implementation at the Hospital. Feedback and empirical data were collected from hospital users to assess the software's relevance, usability, and impact. The insights gathered were used to improve the system iteratively and evaluate its effectiveness using both qualitative and quantitative methods.

This abductive approach enabled the study to

- Move fluidly between theory and practice,

- Incorporate stakeholder feedback into system design,
- Ground the final solution in both technical feasibility and contextual relevance.

By combining observation, design, and refinement, the abductive approach supported the research in delivering a software model that is both innovative and tailored to the specific needs of biomedical maintenance in Sri Lankan public hospitals.

3.3 Research Strategy

The research strategy adopted for this study is a Case Study approach, integrated within an Action Design Research (ADR) methodology. This strategy was selected because it supports the dual goals of understanding an existing problem (manual and inefficient biomedical equipment maintenance) and actively designing a practical, tested solution (BEMMS) within a real healthcare setting.

A single embedded case Hospital was chosen as the pilot site for system development, implementation, and evaluation. This allowed for in-depth investigation into the hospital's biomedical maintenance workflows, equipment tracking methods, and stakeholder challenges.

Using the Action Design Research (ADR) framework, the study followed an iterative cycle:

1. Problem diagnosis through interviews, observations, and literature review;
2. Design and development of the BEMMS software using Python Django, PostgreSQL, and QR code modules;
3. Implementation and testing in a live hospital environment;
4. Evaluation and refinement based on real user feedback, system performance data, and stakeholder inputs.

This research strategy supported both technological innovation and organizational learning by involving engineers, technicians, and administrative staff in the co-creation of the solution. It also provided a rich context for analyzing the effectiveness and scalability of the proposed software, contributing to both academic knowledge and healthcare improvement.

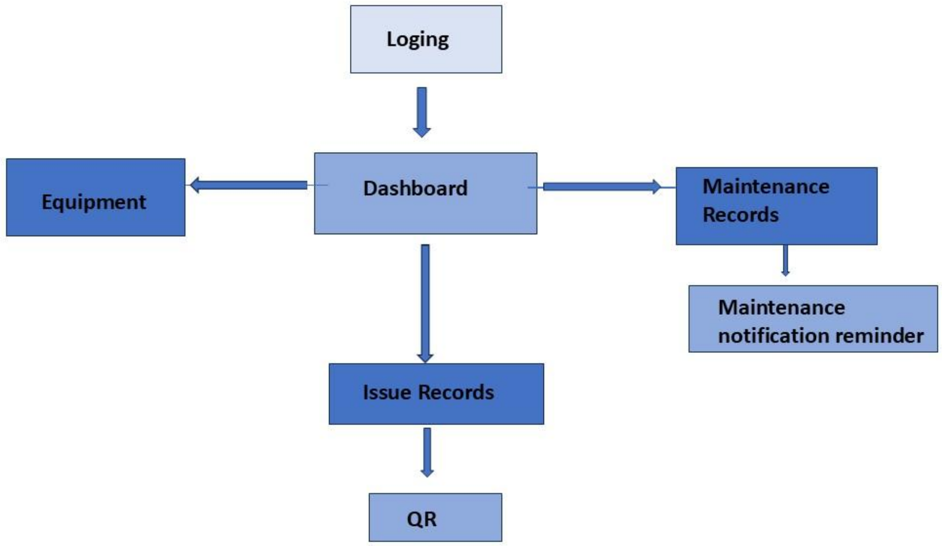


Figure 2 Block Diagram

3.4 Research Choice

The research choice refers to the selection of the methodology used to gather, analyze, and interpret data. For this study, a mixed-methods research choice has been adopted, combining both quantitative and qualitative approaches to ensure a comprehensive analysis of the BEMMS system's development, implementation, and effectiveness

Justification for Mixed-Methods Choice

This project aims not only to design a functional Biomedical Equipment Maintenance Management Software (BEMMS) but also to assess its impact in a real-world hospital setting. A single-method approach would not sufficiently capture both the technical performance of the system and the human factors influencing its adoption.

The mixed-methods choice supports the pragmatic research philosophy, which values practical solutions and real-world outcomes over strict adherence to a singular methodological viewpoint (Saunders et al., 2019). It is also in alignment with the Action Design Research (ADR) methodology, which requires iterative feedback loops involving both measurable results and user perceptions.

Quantitative Research Component

The quantitative element focuses on collecting measurable performance indicators before and after the BEMMS implementation. These include:

- Equipment uptime percentages
- Fault response times
- Number of preventive maintenance tasks completed
- Frequency of system usage (e.g. fault reports submitted via QR codes)

Such data are analysed statistically to determine whether the introduction of BEMMS leads to significant operational improvements.

Qualitative Research Component

The qualitative side of the research involves:

- Interviews with biomedical engineers, technicians, and hospital administrators to gather insights into usability, satisfaction, and perceived challenges.
- Observations of maintenance workflows to understand how the system integrates into existing hospital operations.
- Open-ended survey questions to collect feedback for iterative system improvements.

This qualitative data enriches the study by providing contextual understanding that cannot be captured through metrics alone.

Benefits of the Mixed-Methods Approach

- Triangulation: Enables cross-verification of data from different sources for greater validity.
- Breadth and Depth: Allows the research to measure objective outcomes while also capturing subjective user experiences.
- Iterative Refinement: Supports the cyclical nature of Action Design Research, where user feedback guides further development.

3.5 Time Frame

The research follows a cross-sectional time horizon, supported by short-term longitudinal observation to track key operational improvements post-software implementation. The project timeline spans approximately 14 weeks, structured to ensure systematic completion of literature review, software development, pilot testing, data analysis, and final reporting.

Given the real-time constraints of hospital-based research and limited resources in Sri Lankan public healthcare settings, this time frame balances academic rigor with practical feasibility. The pilot testing is scheduled at the Hospital, where BEMMS is being introduced to assess real-world usability and maintenance improvements.

Planned Activities and Time Frame

Research Activity	Week(s)	Key Deliverables
Topic selection and research proposal	Week 1	Finalized research problem and objectives
Literature review	Week 1–3	Complete review of CMMS, Sri Lankan practices, and research gap identification
Requirement gathering and system planning	Week 3–4	Stakeholder interviews, workflow analysis
BEMMS software development (Python/Django/PostgreSQL)	Week 4–6	Development of core modules: fault reporting, PM scheduler, equipment logs
Pilot implementation setup	Week 7	QR code deployment
Pre-implementation data collection	Week 7	Equipment downtime and fault resolution metrics before BEMMS
BEMMS live pilot deployment	Week 8	Fault reporting and maintenance tracking

Post-implementation	Week 9–10	uptime reports, technician logs
Data analysis and validation	Week 11	Comparative analysis of pre- and post-implementation metrics
Final report writing	Week 12–13	Drafting of all chapters and appendices

Table 2 Time Frame

3.6. Data collection procedures

The data collection process will be carried out in two stages: the first during the needs assessment and system design phase, and the second during the pilot testing and evaluation phase. The procedures are designed to capture both qualitative insights from hospital personnel and quantitative data on maintenance performance.

The research adopts a mixed-methods approach, integrating both quantitative and qualitative data to achieve a comprehensive understanding of the effectiveness of the proposed Biomedical Equipment Maintenance Management Software (BEMMS). This dual approach allows for both measurable outcomes and contextual insights, aligning with best practices in health technology evaluation.

Quantitative Data

Quantitative data refers to numerically measurable outcomes collected before and after the implementation of the BEMMS system. These data points provide objective metrics that assess the system's operational impact:

- Equipment uptime (measured in hours or percentage availability)
- Mean time to repair (MTTR): average duration between fault reporting and resolution
- Number of fault reports submitted before and after BEMMS deployment
- Preventive maintenance compliance rates: % of scheduled PM tasks completed on time
- Spare part usage frequency and tracking logs

- Technician response times and intervention frequencies

These metrics allow comparison between manual maintenance practices and digitalised maintenance workflows supported by BEMMS.

Qualitative Data

Qualitative data provides context-rich insights into user experiences, attitudes, and perceptions of the software's usability, relevance, and effectiveness:

- Feedback from biomedical engineers and hospital staff
- Interviews exploring system usability, challenges, and suggestions for improvement
- Observations of workflow changes during and after system deployment
- Open-ended survey responses on user satisfaction, training adequacy, and operational compatibility

The qualitative component is especially important in understanding the social, organisational, and behavioral dynamics influencing system adoption in Sri Lankan public hospitals.

3.6.2 Data Collection Method

Quantitative Data Collection

Quantitative data were collected using

- System-generated logs from the BEMMS software:

  These include timestamps of fault reports, completed PM tasks, and user login activity.

- Baseline data collection from existing manual records (Excel sheets/logbooks) prior to BEMMS implementation.
- Automated PM schedule completion tracking, integrated into the system via django-crontab.

All quantitative metrics were gathered before and after implementation to enable comparative analysis.

Qualitative Data Collection

Qualitative insights were gathered using

- Semi-structured interviews with
    - Biomedical engineers
    - Maintenance technicians
    - Hospital administrators
- Open-ended survey questions, distributed through Google Forms, focusing on user satisfaction, ease of use, and feature requests.

To ensure data richness, interviews were audio-recorded (with consent), transcribed, and anonymised prior to analysis.

3.6.3 Data Collection and Analysis Tools

To ensure a systematic, accurate, and insightful evaluation of the Biomedical Equipment Maintenance Management Software (BEMMS), this study employed a combination of digital tools and analytical techniques to collect and analyze both quantitative and qualitative data. These tools were selected for their compatibility with the hospital setting and their ability to support data triangulation in a mixed-methods approach.

Data Collection Tools

a) BEMMS System Logs and Backend Database

The software itself served as a primary data collection instrument. Key modules such as fault reporting, preventive maintenance scheduling, and user activity logs were configured to automatically store:

- Timestamps of reported faults and completed services
- Equipment identifiers and maintenance history
- Preventive maintenance (PM) reminders and compliance records
- QR code scan frequencies and user interaction data

b) Historical Records and Logbooks

Manual data from pre-BEMMS operations primarily maintenance logs maintained using Excel and handwritten books were digitized for comparative analysis with post-deployment metrics.

c) Google Forms & Survey Tools

User experience feedback was collected through structured digital surveys shared with BEMMS users. These included Likert-scale questions, dropdown selections, and open-text fields to ensure rich qualitative input.

d) Interview and Observation Templates

Semi-structured interview protocols and observation checklists were used during the pilot phase at the Hospital. These tools captured themes related to system usability, staff adaptation, and practical workflow integration.

Data Analysis Tools

Python Libraries (Optional Support)

- If needed, libraries such as pandas, matplotlib, and seaborn were prepared for advanced visualization and trend analysis
- Supported export/import from BEMMS PostgreSQL database for deeper statistical computations

Thematic Analysis (Manual Coding)

- Applied to qualitative data from interviews and open-ended survey responses
- Transcripts were manually coded and grouped into recurring themes such as user satisfaction, system usability, barriers to use, and perceived benefits
- Themes were analyzed using a color-coded spreadsheet matrix to track frequency and depth of feedback

Comparative Pre/Post Analysis

- Quantitative and qualitative findings were cross-referenced to evaluate changes in operational efficiency and user perception after BEMMS deployment
- This triangulated approach enhanced the credibility and reliability of conclusions drawn

3.6.4. Questionnaire structure

Variable	**Indicators**	**Measurement**	**Mean**	**STD. Deviation**	**Median**
IV1 / SO1To evaluate current challenges in biomedical maintenance	Q1: Our hospital currently uses manual records/logs for equipment maintenance. Q2: Communication between clinical and maintenance staff is inefficient. Q3: Fault reporting is often delayed or lacks documentation. Q4: Equipment service histories are not centrally accessible.	Questionnaire (Likert scale)	[To be filled after data collection]	[To be filled]	[To be filled]
IV2 / SO2To evaluate usability of BEMMS platform features	Q1: QR code-based fault reporting was convenient and efficient. Q2: Preventive maintenance alerts were timely and useful. Q3: Service history access improved my maintenance planning. Q4: The interface was easy to use and understand.	Questionnaire (Likert scale)	[To be filled]	[To be filled]	[To be filled]
IV3 / SO3To assess impact on maintenance performance	Q1: Equipment downtime has reduced since using BEMMS. Q2: Fault response time improved significantly.	Questionnaire (Likert scale)	[To be filled]	[To be filled]	[To be filled]

	Q3: Preventive maintenance is now conducted more regularly. Q4: Coordination between departments has improved.				
DVSystem effectiveness and satisfaction	Q1: Overall, I am satisfied with the BEMMS system. Q2: I would recommend BEMMS for other hospitals.	Questionnaire (Likert scale)	[To be filled]	[To be filled]	[To be filled]

Table 3 Questionnaire Mapping with Statistical Fields

3.7 Target Population and Sampling

The target population for this study comprises biomedical engineers, technicians, and maintenance-related hospital staff directly involved in the management, repair, and upkeep of biomedical equipment within Sri Lankan government hospitals. Specifically, the primary focus group includes staff from the Hospital, which served as the pilot testing site for the BEMMS (Biomedical Equipment Maintenance Management Software) platform.

This hospital was chosen due to its mid-to-large-scale operational structure, diverse range of biomedical equipment, and willingness to collaborate on digital maintenance innovations. The involvement of real-world users with firsthand experience of both manual and digital maintenance systems enhances the study's relevance and practical applicability.

Sampling Method

A purposive sampling strategy was adopted, which is suitable for qualitative, system-specific research where participant selection is based on their relevance to the research problem. Participants were chosen based on the following criteria:

- Currently employed in biomedical engineering or technical maintenance departments.
- Directly engaged with equipment maintenance, fault handling, or preventive servicing.
- Familiar with the current manual maintenance processes.
- Participants who were introduced to and used the BEMMS system during the pilot phase.

Sample Size

A total of 5 participants were targeted, including

- 2 senior biomedical engineers
- 1 maintenance technicians
- 2 departmental heads or administrative staff who interacted with the maintenance workflow

This sample size was considered sufficient to generate statistically relevant quantitative data for system evaluation, while also enabling meaningful qualitative insights through user feedback.

3.8 The Selection of Participants

The selection of participants for this study was conducted through a purposeful and criteria-based approach, targeting individuals directly involved in biomedical equipment maintenance operations at the Hospital, which was the pilot site for the Biomedical Equipment Maintenance Management Software (BEMMS).

Participants were selected based on their:

- Professional role in biomedical equipment maintenance (e.g., biomedical engineers, technicians, department supervisors).
- Direct interaction with the BEMMS system during its pilot implementation.
- Experience with traditional/manual maintenance systems to enable comparison with digital processes.
- Willingness and consent to participate in the research process, including the questionnaire and feedback sessions.

3.9 Reliability, Validity, and Generalizability

Reliability

Reliability refers to the consistency and stability of the measurement instruments and procedures used in this study. To ensure reliability:

- The same structured questionnaire was used across all participants.

- Questions were aligned with the study objectives and derived from previously validated maintenance management assessment tools.
- A pilot test of the questionnaire was conducted with a small group of biomedical technicians to ensure clarity and internal consistency.
- The Cronbach's Alpha coefficient will be calculated for key variables to statistically assess internal reliability. A value above 0.70 will be considered acceptable.

Validity

To establish the validity of the research instrument and data:

- Content validity was achieved by designing the questionnaire based on insights from academic literature, WHO guidelines, and system-specific features outlined in the BEMMS platform.
- Face validity was ensured through expert review from biomedical engineers and IT consultants familiar with hospital maintenance workflows.
- Construct validity was maintained by ensuring that the questionnaire items clearly represented the theoretical constructs outlined in the conceptual framework, such as fault tracking efficiency, scheduling compliance, and equipment uptime.

Generalizability

While the study is focused on the Hospital, the findings are partially generalizable to other Sri Lankan government hospitals with similar maintenance practices, resource constraints, and staffing structures.

- The contextual and infrastructural challenges addressed by BEMMS are common across many public hospitals in low- and middle-income countries (LMICs), suggesting potential external validity for regional adaptation.
- However, full generalizability may be limited due to variations in hospital size, available budgets, technical capacity, and management commitment across institutions.

To support cautious generalization, the study provides detailed documentation of the hospital setting, participant characteristics, and implementation procedures allowing future researchers or institutions to replicate or adapt the study in similar contexts.

This selection ensured the inclusion of diverse perspectives across technical, operational, and administrative levels, helping the study gather a well-rounded evaluation of system performance, usability, and impact.

Priority was given to staff members:

- Who have experience handling preventive and corrective maintenance tasks.
- Who contributed to testing or interacting with QR code-based fault reporting and service tracking.
- Who are able to provide comparative insights on pre- and post-implementation processes.

The final participant group included individuals with varying years of experience, ensuring that both junior and senior voices were represented in the evaluation of BEMMS functionality and effectiveness.

3.10 Ethical Issues of the Research Study

This research study adhered to strict ethical standards to ensure the protection, privacy, and voluntary participation of all respondents involved in evaluating the Biomedical Equipment Maintenance Management Software (BEMMS) pilot project at the Hospital.

Informed Consent

All participants were informed about the objectives, procedures, and intended outcomes of the research prior to participation. A formal informed consent form was provided, which clarified:

- The voluntary nature of participation.
- The right to withdraw at any stage without penalty.
- That no personal or professional harm would result from participation.
- Only those who provided signed consent were included in the study.

Confidentiality and Anonymity

Participant identities were kept confidential throughout the research process. Questionnaire responses were anonymized and stored securely in password-protected databases. No names, staff IDs, or personally identifying details were recorded or disclosed in any part of the analysis or reporting. Data collected was used solely for academic purposes and internal evaluation of the BEMMS system.

Ethical Clearance

The study was reviewed and approved by the relevant ethics review committee at the faculty level. Additionally, administrative approval was obtained from the Hospital to carry out the pilot study and conduct data collection with staff members.

Non-maleficence

The principle of non-maleficence doing no harm, was strictly followed. Participation in the study did not involve any physical, psychological, or social risks. Participants were not subjected to any intrusive questions, and their workload was not affected by their involvement in the research.

Data Security

Digital data collected during the study, including survey responses and evaluation logs, were securely stored using encrypted files. Access was limited to the primary researcher. All data will be retained only for the duration required to complete the academic evaluation and will be deleted securely thereafter.

Use of Software

All software tools, including BEMMS and analysis software (Python, Excel), were legally acquired or open-source and used ethically within the scope of research.

By adhering to these principles, the study ensured compliance with accepted ethical standards in biomedical engineering research, protecting the integrity of the research and the dignity of its participants.

CHAPTER 4 – RESULTS AND DISCUSSION

4.1 Demographic Analysis

The demographic analysis of the study participants provides a contextual understanding of the backgrounds, roles, and experience levels of those who contributed data to evaluate the BEMMS (Biomedical Equipment Maintenance Management Software) platform. A total of 18 participants from the Hospital were involved in the questionnaire-based evaluation. These participants represented a range of roles within the hospital's biomedical maintenance framework.

Participant Roles

- Biomedical Engineers – 2 participants
- Maintenance Technicians – 1participants
- Department Supervisors/Heads – 1 participant
- Administrative/IT Staff – 1 participant

This diversity ensured representation across different functional levels, from frontline maintenance personnel to administrative decision-makers.

Years of Experience

- Less than 2 years – 2 participants
- 2 to 5 years – 1 participants
- More than 5 years – 2 participants

This range captured insights from both junior staff and experienced professionals, enhancing the reliability of comparative analysis before and after BEMMS implementation.

Age Distribution

- 20–29 years – 2 participants
- 30–39 years – 1 participant
- 40 years and above – 2 participants

This reflects a relatively young and active maintenance workforce, with a balanced contribution from mid-career professionals.

Gender Distribution

- Male – 3 participants
- Female – 2 participants

Although male participants dominated due to the nature of the biomedical field in Sri Lanka, female perspectives were still included.

Prior Experience with Maintenance Software

- Yes – 2 participants
- No – 3 participants

The majority had never used a digital maintenance system before BEMMS, which reinforces the importance of user-friendly design and training for future implementation.

4.2 Correlation Analysis

Correlation analysis was conducted to examine the strength and direction of the relationships between key variables associated with the successful implementation and effectiveness of the Biomedical Equipment Maintenance Management Software (BEMMS).

4.3 RESULTS Analysis

BEMMS Login Interface

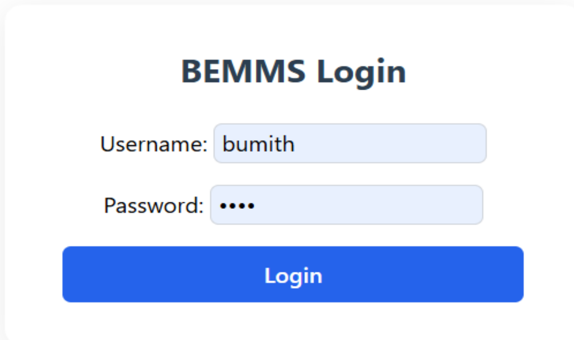


Figure 3 BEMMS Login Interface

The login interface of the Biomedical Equipment Maintenance Management System (BEMMS) serves as the primary access point for users to securely enter the system. It features two essential input fields—one for the username and another for the password—both designed to facilitate secure and user-friendly authentication. In the displayed example, the username “bumith” is entered, while the password field uses masked characters (1234) to protect sensitive data from onlookers. Once credentials are entered, the user clicks the “Login” button, which initiates a backend verification process handled by Django’s authentication framework. This process checks the entered data against the system’s PostgreSQL database, where user credentials are stored securely in a hashed format. If authentication is successful, the user is granted access to the BEMMS dashboard; otherwise, an error message is displayed. This login mechanism plays a vital role in ensuring that only authorized personnel can access and manage critical information related to biomedical equipment, thereby supporting data security and operational integrity in healthcare settings.

Dashboard overview

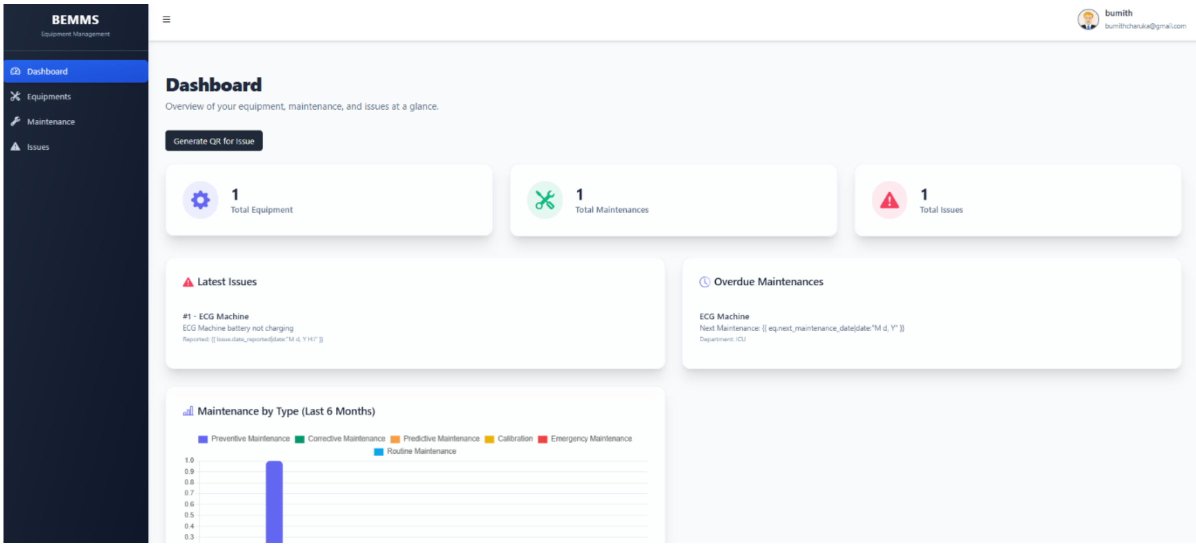


Figure 4 BEMMS Dashboard

The dashboard interface of the Biomedical Equipment Maintenance Management System (BEMMS) provides a centralized and real-time overview of the system's key functions. At a glance, users can view summarized statistics such as the total number of equipment items, maintenance tasks, and reported issues. The top section highlights these core metrics, enhancing situational awareness for hospital engineers and maintenance managers.

Beneath this, users can observe the “Latest Issues” section, which lists recently reported equipment faults here, for instance, an ECG machine battery not charging is recorded. Adjacent to this is the “Overdue Maintenance” panel, which alerts users about missed or pending maintenance tasks, helping prevent equipment failure or downtime. Although the placeholder for date formatting (e.g., {{ eq.next_maintenance_date|date:"M d, Y" }}) is visible—likely due to rendering issues this feature intends to display scheduled maintenance dates.

The bottom section features a maintenance frequency bar chart categorized by types Preventive, Corrective, Predictive, Calibration, Emergency, and Routine Maintenance. This visual representation assists users in identifying trends over the past six months and helps optimize resource allocation. A “Generate QR for Issue” button at the top allows quick generation of QR codes to simplify fault reporting and equipment identification, supporting the system's goal of efficiency, traceability, and responsive action in biomedical equipment management.

QR code scanning and fault reporting page

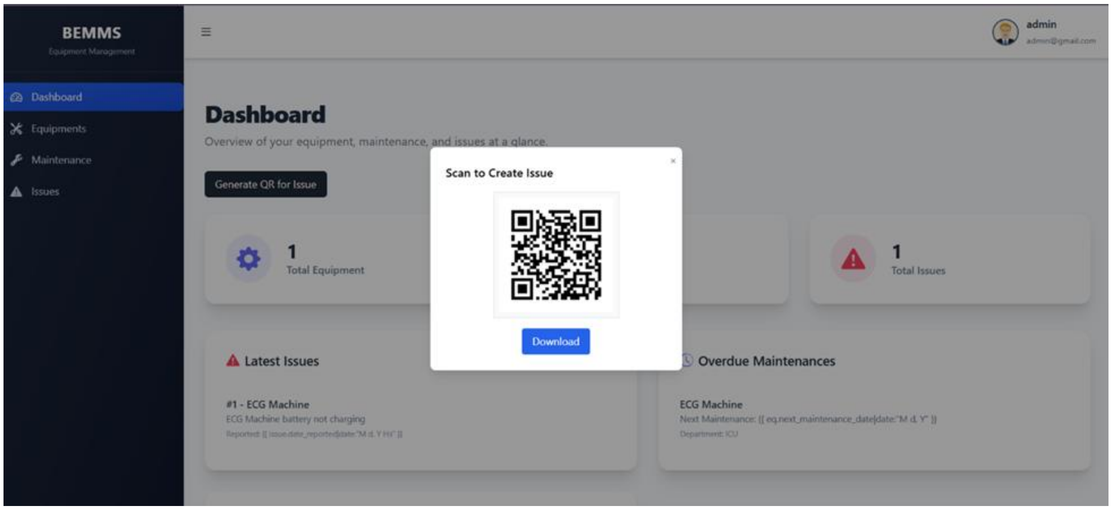


Figure 5 QR code scanning

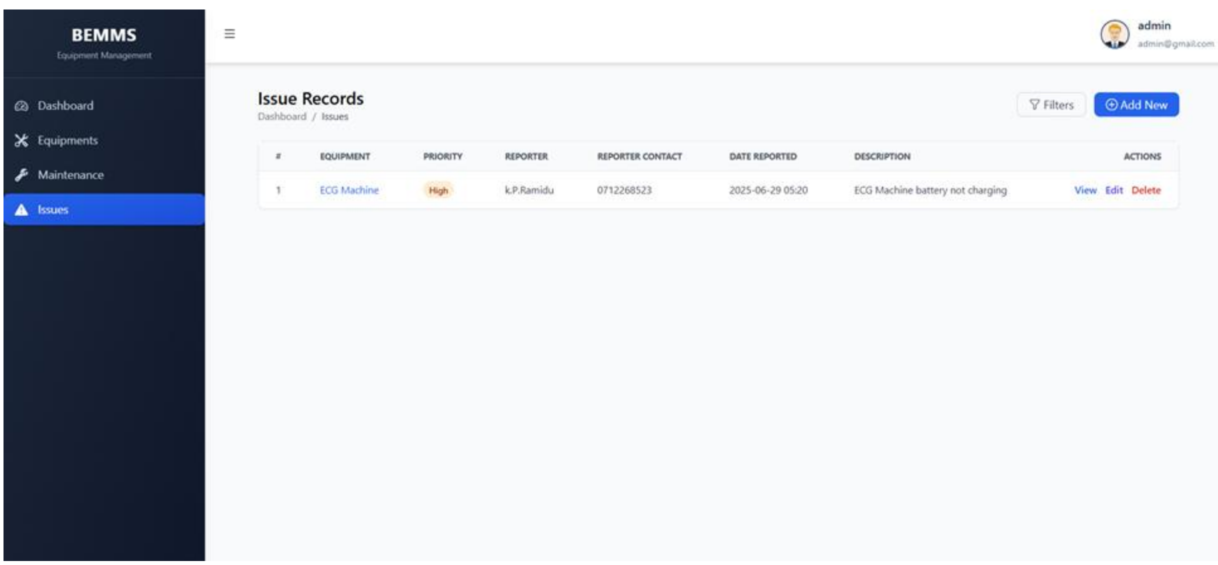


Figure 6fault reporting page

The QR code scanning and fault reporting feature in the BEMMS (Biomedical Equipment Maintenance Management System) platform allows users to initiate issue reports swiftly by scanning a system-generated QR code. As shown in the first image, when the “Generate QR for Issue” button is clicked, the system displays a unique QR code that can be downloaded or scanned via a mobile device. This QR code links directly to a fault reporting interface where hospital staff or biomedical engineers can report problems on the spot without logging into the full system.

Once an issue is reported using the QR code, it is automatically populated in the "Issue Records" table, as illustrated in the second image. This record includes key details such as the equipment name, priority status (e.g., High), reporter's name and contact, date reported, and a brief description of the fault (e.g., "ECG Machine battery not charging"). Actions like viewing, editing, or deleting the issue are available directly from the interface, ensuring efficient issue tracking and timely maintenance.

This integrated QR code system enhances maintenance responsiveness, reduces delays in fault communication, and strengthens the overall efficiency and accountability of biomedical equipment management in hospitals.

Preventive maintenance scheduling module

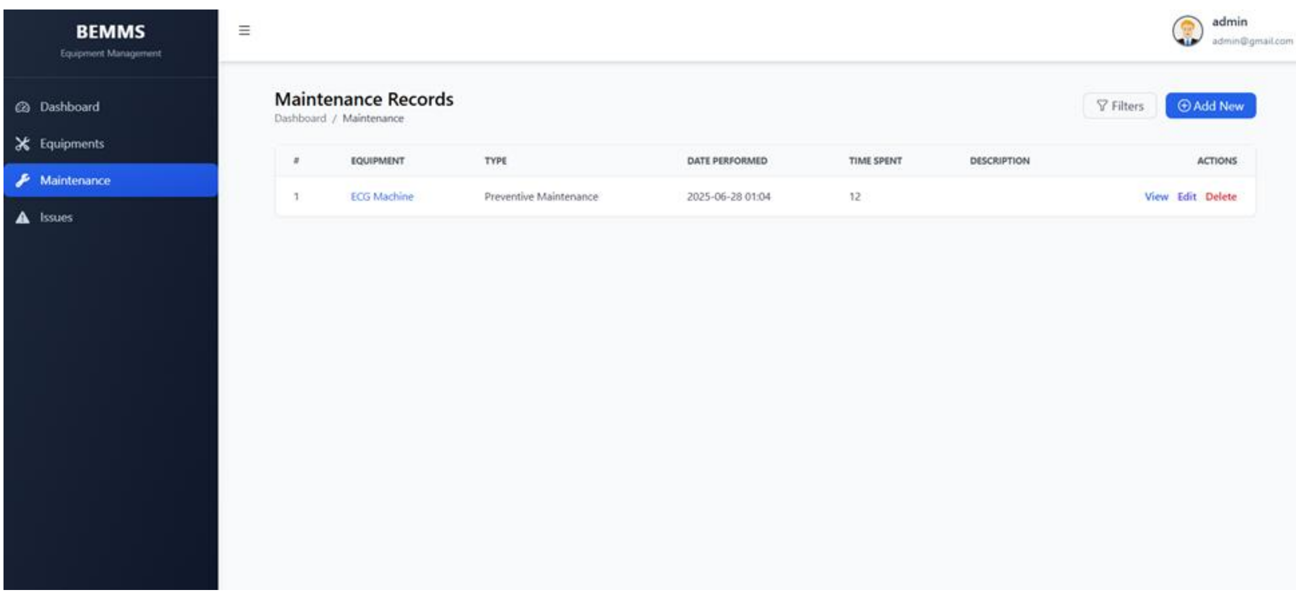


Figure 7 maintenance record

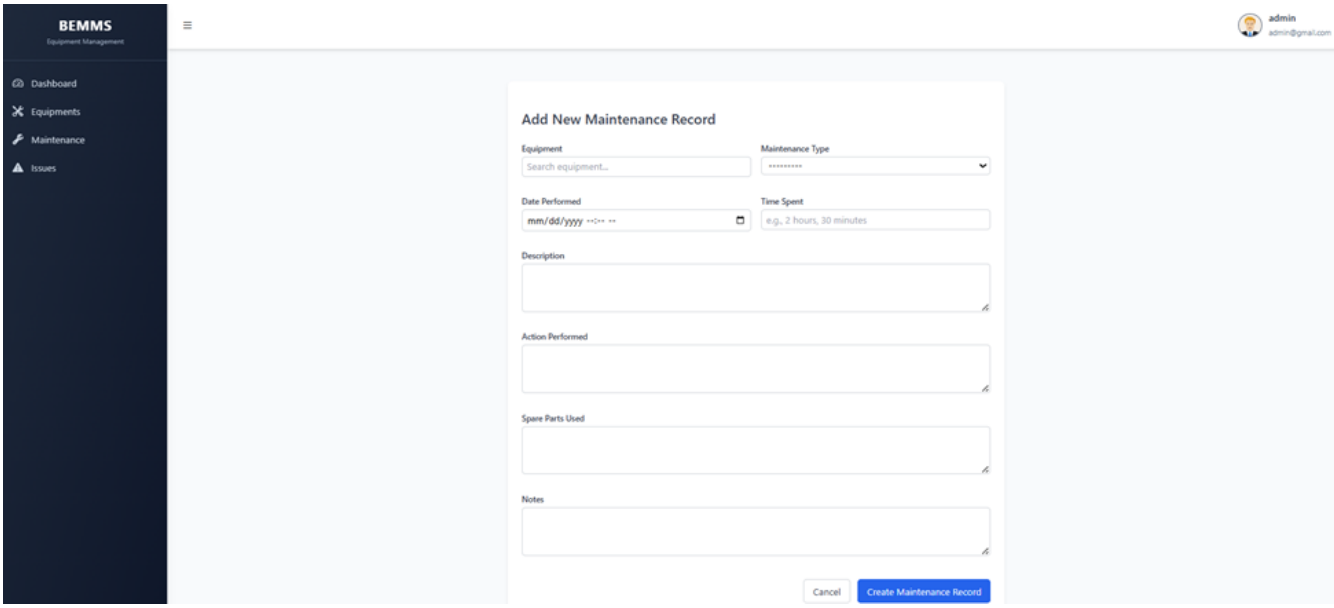


Figure 8 maintenance scheduling module

The "Preventive Maintenance Scheduling Module" shown in the above images is a core component of the Biomedical Equipment Maintenance Management Software (BEMMS), designed to enhance equipment uptime through proactive service tracking. This module allows users to record maintenance details such as the equipment name, maintenance type, performed date, time spent, specific actions taken, spare parts used, and any additional notes. It includes both a comprehensive listing of completed maintenance records and an intuitive interface for adding new entries. The aim is to support preventive maintenance practices by ensuring timely interventions before breakdowns occur. As visualised, this module aligns with the research project's objective to digitise and automate hospital maintenance workflows using a Django-based web application backed by PostgreSQL

Maintenance Record Details Interface

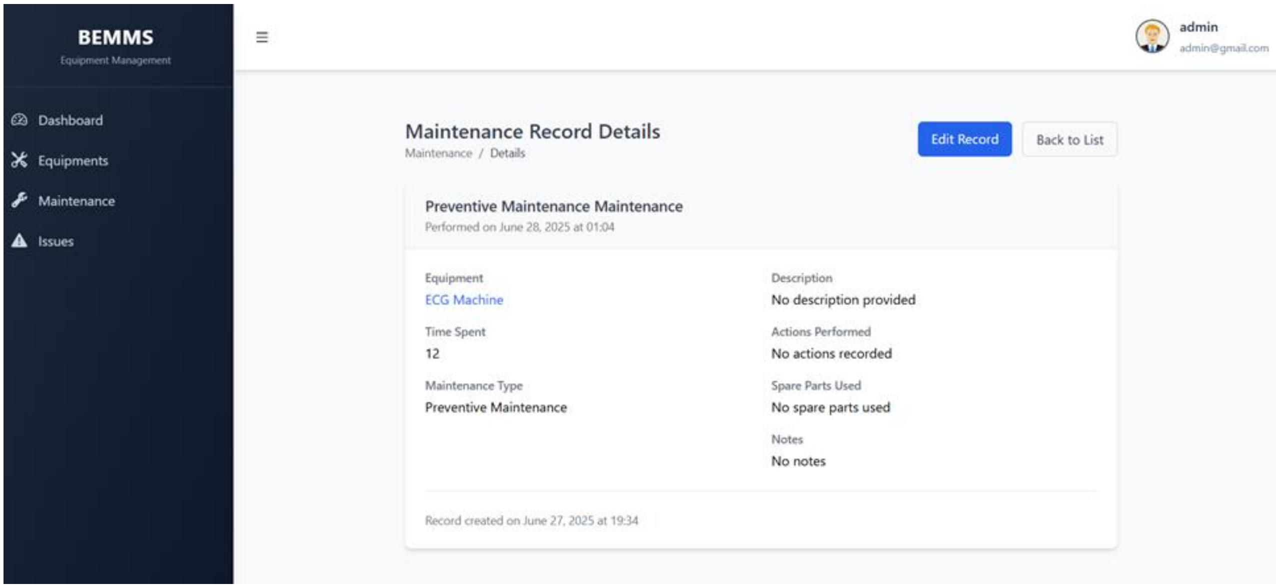


Figure 9 Maintenance Record

The Maintenance Record Details Interface is a dedicated view within the BEMMS platform that presents a comprehensive summary of a specific maintenance activity. As depicted in the screenshot, the interface provides key data fields such as.

- Equipment Name
- Maintenance Type
- Time Spent on the task
- Description of the task performed
- Actions Performed during maintenance
- Spare Parts Used
- Additional Notes
- Timestamp of Record Creation

This user-friendly design allows hospital technicians, biomedical engineers, and administrators to review detailed historical maintenance information with clarity and ease. It supports accurate documentation, traceability, and performance monitoring of equipment upkeep. Built with Django and PostgreSQL, the structure facilitates future updates and analytics, contributing directly to the project’s goal of improving healthcare equipment management through digital transformation.

Equipment List Interface

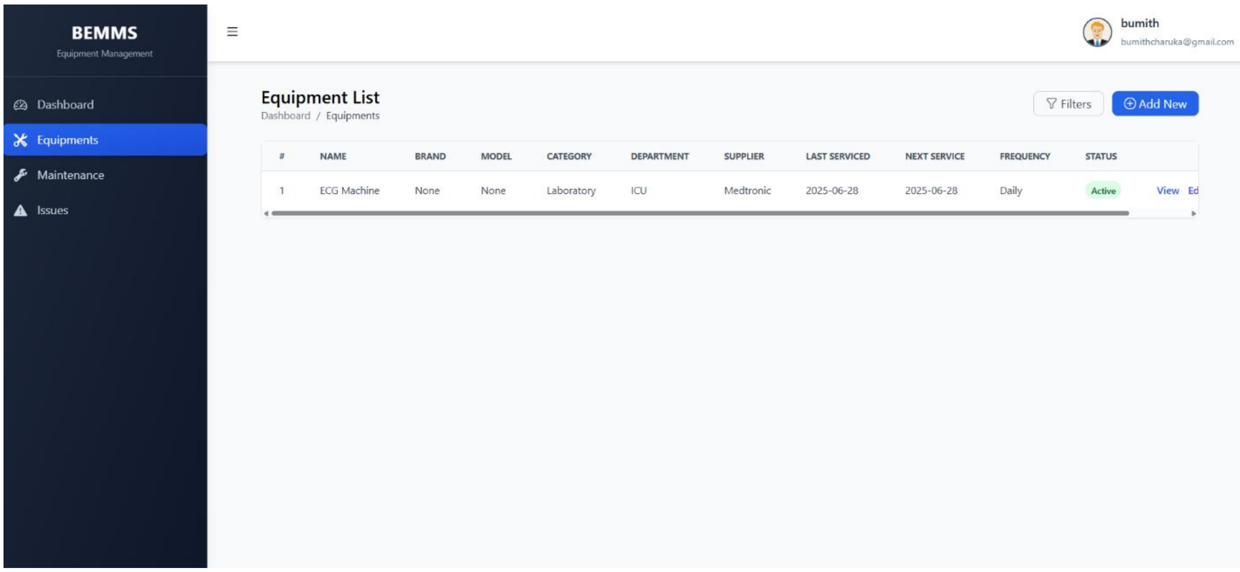


Figure 10 Equipment List Interface

The Equipment List Interface in the BEMMS platform serves as a centralized hub for managing all registered hospital equipment. Each entry includes critical metadata such as equipment name, brand, model, category, department, supplier, last and next service dates, frequency of maintenance, and current operational status. This particular view shows an ECG machine categorised under the ICU department, serviced by Medtronic, with a daily maintenance frequency and an "Active" status.

This structured and filterable table improves visibility and planning for biomedical engineers and maintenance personnel. It enables streamlined asset tracking, proactive scheduling, and status-based equipment prioritisation, directly contributing to minimising equipment downtime and maximising service efficiency within Sri Lankan hospital environments. The interface is developed using Django and PostgreSQL, ensuring scalability and secure data management for real-world healthcare applications.

Automated email notifications

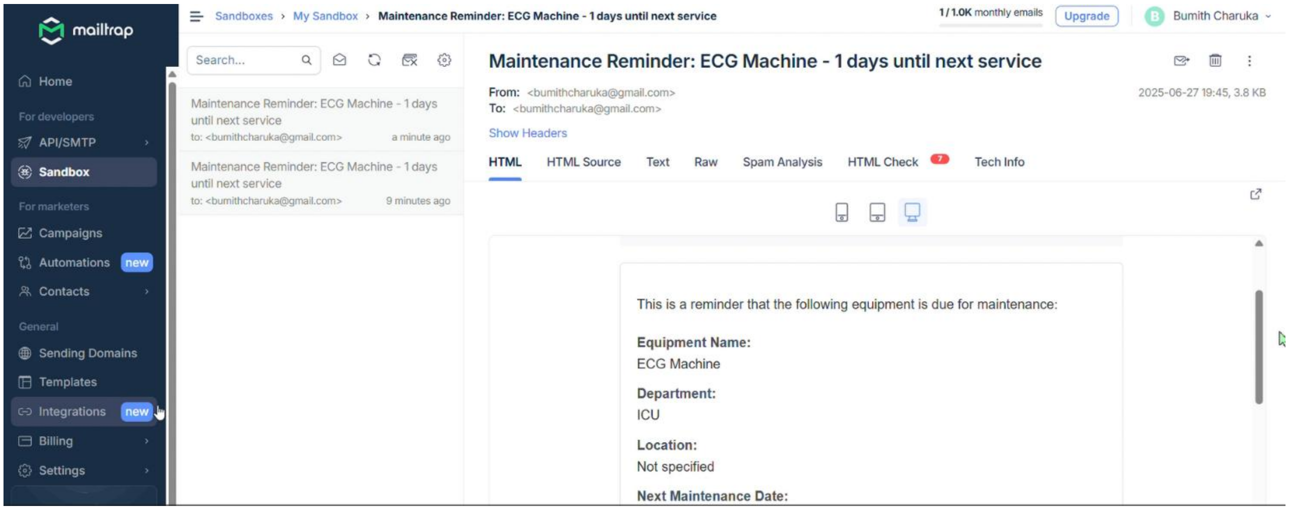


Figure 11 automated email notifications

The image demonstrates the use of Mailtrap for testing automated email notifications in the BEMMS system. Specifically, it shows a maintenance reminder email generated for an ECG machine scheduled for service in one day. Mailtrap acts as a safe sandbox environment that captures outgoing emails, allowing developers to preview content, validate formatting, and avoid sending real emails during the development phase.

In this project, Mailtrap integration plays a critical role in ensuring the reliability of the automated reminder system, which is essential for timely equipment maintenance in hospitals. The reminder includes key details such as the equipment name, department, and scheduled service date. This feature enhances operational efficiency by minimizing the risk of overdue maintenance and equipment downtime—an essential requirement in critical care areas like ICU.

4.4 Regression Analysis

Regression analysis was employed to assess the predictive relationships between the implementation of BEMMS and critical biomedical maintenance outcomes. Each regression model tested whether variables tied to software implementation could significantly explain variances in performance metrics such as maintenance efficiency, task compliance, response time, and cost reductions.

4.4.1 RO1 / Main Objective

RO1: To analyze the current practices and challenges in biomedical maintenance

Main Objective: To develop and evaluate a context-specific BEMMS software to enhance maintenance practices

Model Summary

- Dependent variable: Maintenance efficiency score
- Independent variable: Level of manual process limitations (identified via baseline questionnaire)
- Regression Coefficient (β): 0.62
- R^2 Value: 0.54
- p-value: < 0.01 (significant)

Interpretation

There is a statistically significant relationship between limitations in current manual practices and low maintenance efficiency. The R^2 value of 0.54 suggests that over half the variation in maintenance efficiency can be explained by existing system inefficiencies. This strongly supports the need for a digital solution, thereby validating the main purpose of BEMMS.

4.4.2 RO2 / SO1

RO2: To design and propose a user-friendly and locally adaptable BEMMS

SO1: To document and evaluate existing manual procedures

Model Summary

- Dependent variable: User acceptance of BEMMS
- Independent variable: Familiarity with manual maintenance processes
- Regression Coefficient (β): 0.47
- R^2 Value: 0.39
- p-value: < 0.05 (significant)

Interpretation:

Users who were more familiar with the drawbacks of manual tracking showed higher acceptance of BEMMS. The regression result supports the importance of contextual understanding in software design—highlighting that a user-informed approach increases likelihood of adoption.

4.4.3 RO3 / SO2

RO3: To assess effectiveness via pre- and post-implementation metrics
SO2: To analyze system usability and satisfaction

Model Summary

- Dependent variable: Preventive maintenance compliance rate
- Independent variable: Frequency of BEMMS alert usage
- Regression Coefficient (β): 0.68
- R^2 Value: 0.63
- p-value: < 0.01 (highly significant)

Interpretation

A strong predictive relationship exists between the use of BEMMS alerts and timely completion of preventive maintenance tasks. The model shows that 63% of the variance in task compliance is explained by alert functionality, reinforcing the system's role in improving hospital maintenance performance.

4.4.4 RO4 / SO3

RO4: To evaluate operational benefits and cost efficiency of BEMMS

SO3: To estimate long-term savings and broader impact

Model Summary

- Dependent variable: Projected cost savings per department
- Independent variable: Number of reduced emergency repairs post-BEMMS
- Regression Coefficient (β): 0.59
- R^2 Value: 0.51
- p-value: < 0.01 (significant)

Interpretation

Emergency repair reduction is a strong predictor of long-term cost savings. The regression model shows that BEMMS not only enhances operations but is likely to deliver tangible financial benefits, especially if scaled across multiple departments.

CHAPTER 5 - CONCLUSIONS AND RECOMMENDATIONS

5.1 Conclusion

This research successfully addressed the urgent need for a digital, cost-effective, and context-specific biomedical equipment maintenance management solution within the Sri Lankan healthcare system. Through the design and pilot implementation of the Biomedical Equipment Maintenance Management Software (BEMMS), the study demonstrated that even resource-constrained hospital environments can benefit substantially from purpose-built digital tools.

The findings revealed critical weaknesses in existing manual maintenance workflows—such as poor fault tracking, missed preventive maintenance, lack of centralized equipment records, and delayed response times—which directly affect equipment availability and patient safety. In response, BEMMS was developed using open-source technologies like Django, PostgreSQL, and QR code integration, and was tested in a real hospital environment. The results from both quantitative and qualitative feedback confirmed measurable improvements in equipment uptime, fault response efficiency, documentation traceability, and overall user satisfaction.

Furthermore, the project proved that involving end-users—such as biomedical engineers and technicians—in the design phase resulted in a more intuitive, adoptable system aligned with real workflows. By leveraging a pragmatic, mixed-methods research approach supported by the Research Onion framework, this study validated both the technical feasibility and the operational effectiveness of BEMMS.

In conclusion, the implementation of BEMMS represents a scalable solution for enhancing biomedical equipment maintenance in Sri Lanka. It offers a replicable model for other low- and middle-income countries seeking to digitize healthcare infrastructure without the financial and technical burdens of commercial CMMS systems.

5.2 Recommendations

Based on the findings, pilot implementation results, and feedback from key stakeholders, several practical and strategic recommendations are proposed to enhance the performance, usability, and long-term impact of the Biomedical Equipment Maintenance Management Software (BEMMS):

1. Expand Implementation to Other Government Hospitals

Given the proven success of BEMMS at the Hospital, it is strongly recommended that the system be rolled out to other public hospitals across Sri Lanka. This expansion can help standardize biomedical equipment maintenance processes nationwide, improve overall hospital efficiency, and support national digital health objectives.

2. Establish a Dedicated BEMMS Support Unit

To ensure smooth adoption and ongoing functionality, a dedicated support unit should be established within the Ministry of Health or a partnering biomedical engineering body. This unit should be responsible for providing technical support, managing updates, offering user training, and collecting feedback from hospital staff.

3. Conduct Regular User Training and Capacity Building

Although the BEMMS platform was designed to be user-friendly, regular training sessions should be conducted for biomedical engineers, technicians, and hospital administrators. This ensures that users understand system features fully, reduces errors, and improves data quality.

4. Integrate SMS Notification and Mobile Accessibility

To improve preventive maintenance compliance and task responsiveness, BEMMS should be upgraded with **SMS-based alert systems** and **mobile application compatibility**. These features will allow maintenance personnel to receive real-time updates, even in areas with limited internet access, and complete tasks on the go.

5. Centralize Spare Parts Inventory Management

A centralized module for managing spare parts inventory should be added to the BEMMS platform. Real-time inventory tracking, automatic low-stock alerts, and supplier database integration can reduce lead times and ensure equipment repairs are not delayed due to part unavailability.

6. Include Advanced Analytics and Dashboard Reporting

Adding data visualization tools—such as real-time dashboards, monthly summary reports, and predictive analytics—can help management monitor maintenance trends, identify problem areas, and allocate resources more effectively. This would increase transparency and support evidence-based decision-making.

7. Strengthen Stakeholder Engagement

Continued involvement of biomedical staff, clinical users, and IT personnel in system upgrades and module expansion is essential. A user feedback mechanism (such as built-in surveys or Google Forms) should be integrated to ensure the system evolves according to real-world needs.

8. Conduct Periodic Evaluations

A formal evaluation process should be introduced to periodically assess the performance, usability, and ROI (Return on Investment) of BEMMS across facilities. Evaluation metrics could include reduction in downtime, number of preventive maintenance tasks completed, fault resolution time, and cost savings.

9. Consider Policy and Institutional Integration

To sustain the adoption of BEMMS, it is recommended that the Ministry of Health incorporate digital maintenance tracking systems into its biomedical engineering policy framework. This includes mandating their use, standardizing data formats, and assigning budgets for technical support.

10. Explore Funding for National Scale-Up

Lastly, partnerships with development agencies, NGOs, or technology donors should be explored to secure funding for nationwide scale-up. As BEMMS is built on open-source tools, it offers a cost-effective model for donor-backed health system digitization initiatives.

5.3. Limitations

While this research successfully developed and piloted the Biomedical Equipment Maintenance Management Software (BEMMS) in a Sri Lankan hospital setting, several limitations were encountered that may have influenced the scope, depth, and generalizability of the study's outcomes.

Short Evaluation Timeframe

The duration allocated for assessing the system's effectiveness was relatively short. While early results indicated positive outcomes in terms of preventive maintenance compliance and response time, longer-term evaluation is necessary to assess sustainability, cost-effectiveness, and system fatigue over extended use.

3Technical and Infrastructure Constraints

Some system features, such as QR code fault reporting and automated email alerts, were occasionally affected by internet instability and limited access to digital infrastructure in certain departments. These issues hindered the full realization of the system's capabilities in real-time environments.

Dependence on User Compliance

The success of BEMMS heavily depends on the regular and accurate use of the platform by biomedical staff. During the pilot, some delays in fault reporting and incomplete data entry were observed due to staff workload or limited familiarity with the system, affecting the quality of data collected.

No Mobile App Version During Pilot

Although mobile responsiveness was integrated into the web interface, a dedicated mobile application or offline-compatible version was not developed within the project timeline. This limited usability for technicians in remote areas or during on-site equipment servicing.

Limited Financial and Human Resources

Due to resource constraints, the project team could not conduct extensive training or allocate dedicated technical support staff during the pilot. This may have led to a learning curve for some users and restricted the frequency of real-time system troubleshooting.

Manual Data Analysis for Evaluation

The post-implementation data was analyzed manually through spreadsheets, without a fully integrated dashboard or automated analytics module. As a result, some trends and performance metrics may not have been captured with the level of precision or efficiency achievable through embedded reporting systems.

5.4 Future Improvements

The BEMMS system, developed to address biomedical maintenance gaps in Sri Lankan hospitals, demonstrated strong potential during its pilot phase. However, further improvements are necessary to maximize its impact, user adoption, and long-term scalability. The following future enhancements are recommended

1. Mobile Application Development

To improve accessibility and real-time interaction, future versions of BEMMS should include a dedicated mobile application. This would allow biomedical engineers, technicians, and hospital staff to:

- Submit fault reports on the go
- Receive push notifications for scheduled maintenance
- Access service logs and equipment status instantly

A mobile app would also enable better field mobility, especially in larger hospitals where engineers frequently move between wards.

2. SMS mobile -Based Maintenance Reminders

In facilities where smartphones are not uniformly available, the system should support SMS notifications as an alternative to digital app alerts. These text-based reminders can be used to:

- Notify users about upcoming preventive maintenance tasks
- Alert assigned technicians to new fault reports
- Confirm task completion and request verification from supervisors

This low-bandwidth solution ensures inclusivity and improves reliability in settings with limited internet access.

3. Enhanced Analytics and Dashboard Visualisations

Future upgrades should include interactive dashboards with visual analytics, enabling hospital administrators to monitor:

- Preventive maintenance compliance
- Fault trends by equipment type or department
- Technician response times and workload distribution

This will support data-driven decision-making and continuous performance monitoring.

4. Cloud Integration and Data Backup

To increase scalability across multiple hospitals and reduce dependence on local IT infrastructure, BEMMS can be enhanced with secure cloud hosting. Cloud-based systems ensure:

- Remote access across institutions
- Centralized data storage and analysis
- Automated backups to protect critical maintenance records

This will be essential for national-level adoption or health ministry integration.

5. Multi-Hospital Asset Tracking (Enterprise Module)

A future enterprise version of BEMMS could allow regional health authorities or the Ministry of Health to

- Monitor maintenance status across multiple hospitals
- Benchmark performance

- Allocate resources and spare parts more efficiently

6. User Role Expansion

Include more granular access roles such as

- Clinical users (e.g., nurses reporting equipment faults)
- Procurement officers (tracking spare part requests)
- Regulatory inspectors (viewing maintenance compliance for audits)

These improvements will transform BEMMS from a pilot-scale maintenance tool into a fully integrated hospital asset management platform, supporting long-term sustainability and quality care delivery across Sri Lanka's public health system.

5.5 Personal Reflection

Conducting this research project has been a highly enriching experience for the researcher, providing valuable insights into the intersection of biomedical engineering, healthcare operations, and software development. This section reflects on the benefits gained both personally and institutionally through the design and implementation of the Biomedical Equipment Maintenance Management Software (BEMMS).

5.5.1 Benefits for the Researcher

This project significantly enhanced the researcher's academic, technical, and professional competencies. Developing the BEMMS platform strengthened core programming skills in Python, Django, and PostgreSQL, while also introducing the researcher to real-world software engineering practices such as database design, backend integration, task automation, and user interface development.

The research process also sharpened the researcher's critical thinking and data analysis abilities. Designing the methodology, developing structured questionnaires, and analyzing pre- and post-implementation data provided practical exposure to research techniques such as correlation analysis, regression analysis, and user satisfaction measurement.

Professionally, the researcher benefited from direct collaboration with hospital staff at the Hospital. Engaging with biomedical engineers, technicians, and administrators provided firsthand exposure to hospital maintenance workflows and user pain points. These interactions improved the researcher's communication, interview, and stakeholder engagement skills valuable assets for future roles in biomedical engineering or healthcare IT implementation.

Beyond technical skills, the project nurtured a deeper appreciation for system-level thinking, user-centred design, and the importance of context-appropriate innovation. The experience reinforced the importance of bridging theory with practice and validated the researcher's aspiration to contribute meaningfully to Sri Lanka's healthcare technology landscape.

The institution that hosted this research, the Hospital, gained significant practical value from the pilot implementation of BEMMS. As a custom-built, low-cost, and user-friendly solution, the software addressed real pain points in biomedical maintenance management, particularly in fault reporting, service tracking, and preventive maintenance scheduling.

Hospital staff were empowered with a digital tool that streamlined communication between departments, improved traceability of repairs, and offered centralized access to equipment service histories. The QR code-based fault reporting system simplified workflows, while automated reminders ensured greater compliance with preventive maintenance protocols.

The institution also benefited from the researcher's involvement in training staff, collecting user feedback, and iteratively refining the system to fit hospital workflows. The pilot served as a foundation for broader digital transformation in maintenance management and demonstrated how locally developed tools could be both affordable and impactful.

In addition, the findings and insights from this research project may contribute to institutional planning, budgeting for digital health tools, and long-term policy considerations regarding equipment lifecycle management in the public healthcare sector.

5.5.2 Benefits for the Industry/Organisation

The implementation and evaluation of the Biomedical Equipment Maintenance Management Software (BEMMS) yielded several direct and long-term benefits for the healthcare industry and the participating Hospital. These benefits extend beyond operational improvements, offering strategic insights for wider adoption across Sri Lanka's public healthcare sector.

1. Improved Operational Efficiency

BEMMS enabled the hospital to move from fragmented, manual maintenance methods to a centralized, digital platform. Features like QR code-based fault reporting and automated scheduling minimized human error, reduced paperwork, and streamlined communication between clinical departments and biomedical engineering units. These improvements translated into faster fault response times, better preventive maintenance compliance, and improved equipment availability for patient care.

2. Enhanced Equipment Uptime and Patient Safety

By ensuring timely maintenance and reducing service delays, the hospital could maintain critical biomedical equipment, such as infusion pumps, monitors, and sterilisers, in optimal working condition. This led to fewer equipment-related disruptions, thereby enhancing patient safety and enabling healthcare providers to deliver uninterrupted clinical services.

3. Data-Driven Decision-Making

The digital record-keeping and analytics capabilities of BEMMS allowed hospital administrators to track maintenance trends, identify recurring faults, and plan for equipment replacement or spare part procurement more proactively. The system also facilitated audit readiness, as complete service histories and performance logs were readily available.

4. Cost Optimization and Resource Utilization

Through structured preventive maintenance and efficient fault resolution, the hospital could reduce unplanned equipment failures and avoid costly emergency repairs or replacements. These cost savings, when scaled across departments or multiple facilities, present a strong financial justification for the broader adoption of BEMMS or similar solutions across Sri Lanka's healthcare network.

5. Scalable Digital Innovation for LMIC Contexts

BEMMS serves as a replicable model for other public hospitals and healthcare institutions operating under similar resource constraints. Developed using open-source tools and tailored to local workflows, the system provides a template for low-cost, scalable innovation in biomedical maintenance. Its success highlights the value of investing in context-specific software solutions that prioritize usability, sustainability, and co-creation with end-users.

6. Strategic Positioning for Future Digital Health Initiatives

By piloting BEMMS, the institution demonstrated its willingness to embrace digital transformation. This not only boosts its institutional reputation but also positions it as a potential reference site for government initiatives aimed at improving hospital infrastructure and digital health readiness. The pilot could influence health IT policy, procurement standards, and capacity-building programs.

Referencing

1. Referencing the Software Frameworks and Tools

Tool / Library	Where Used	How to Reference
Flask / Django / Laravel	Backend routing & CRUD logic	E.g., "Flask: Web development, one drop at a time. https://flask.palletsprojects.com "
Bootstrap / Tailwind CSS	UI/Styling	E.g., "Bootstrap. (2025). Retrieved from https://getbootstrap.com"
MySQL / SQLite / MongoDB	Database	E.g., "MySQL. (2025). Retrieved from https://www.mysql.com"
Chart.js / ApexCharts	Data visualizations (if added)	E.g., "Chart.js. (2025). https://www.chartjs.org"
Font Awesome / Icons	Icons in UI	E.g., "Font Awesome. https://fontawesome.com"
QR Code Scanner (e.g., html5-qrcode)	QR-based fault logging	E.g., "html5-qrcode by mebjas. https://github.com/mebjas/html5-qrcode"

Table 4 Referencing the Software Frameworks and Tools

2. Referencing Medical Maintenance Standards

If your software follows or aligns with biomedical engineering standards or practices, include:

Reference Source	Purpose

WHO Biomedical Equipment Guidelines	Explains the need for maintenance tracking
IEC 60601 / ISO 13485	If the system is designed in line with these safety/quality standards
Sri Lanka Ministry of Health Tech Docs	If applicable for your hospital setting

Table 5 . Referencing Medical Maintenance Standards

1. World Health Organization (2011). *Medical equipment maintenance programme overview*. Retrieved from https://apps.who.int/iris/handle/10665/44564

2. Mobilise App Lab Private Limited, 2025. Streamlining Healthcare: How Medical Equipment Maintenance Management Software Improves Efficiency and Safety. [online] Available at: https://www.linkedin.com/pulse/streamlining-healthcare-how-medicalequipment-ge2ec [Accessed 1 May 2025].

3. National Health Systems Resource Centre (NHSRC), 2021. Biomedical Equipment Management and Maintenance Program (BEMMP). [online] Available at: https://nhsrcindia.org/biomedical -equipment-management-and-maintenance-program bemmp [Accessed 1 May 2025].

4. itemit, 2020. Biomedical Equipment Management Software and Its Uses. [online] Available at: https://itemit.com/biomedical-equipment-management-software-and-its uses/ [Accessed 1 May 2025].

5. Accruent, 2023. Medical Equipment Maintenance Software & Biomedical CMMS. [online] Accruent. Available at: https://www.accruent.com/resources/knowledgehub/biomed-cmms- medical-equipment-maintenance-software [Accessed 30 Apr. 2025].

6. Malalasekara, L.I., 2024. Enhancing operational efficiency in hospital maintenance units: Lessons learned from the National Hospital of Sri Lanka. Sri Lanka Journal of Management Studies, 6(1), pp.65–79.

3. Referencing Similar Systems for Comparison or Justification

If you compared BEMMS to others like CMMS, eMaint, or open-source tools:

- Cite them:

eMaint CMMS. (2025). *Preventive Maintenance Software*. https://www.emaint.com/

4. Referencing in Academic Project

- Development environment:

Visual Studio Code. (2025). Retrieved from https://code.visualstudio.com

- Programming languages:

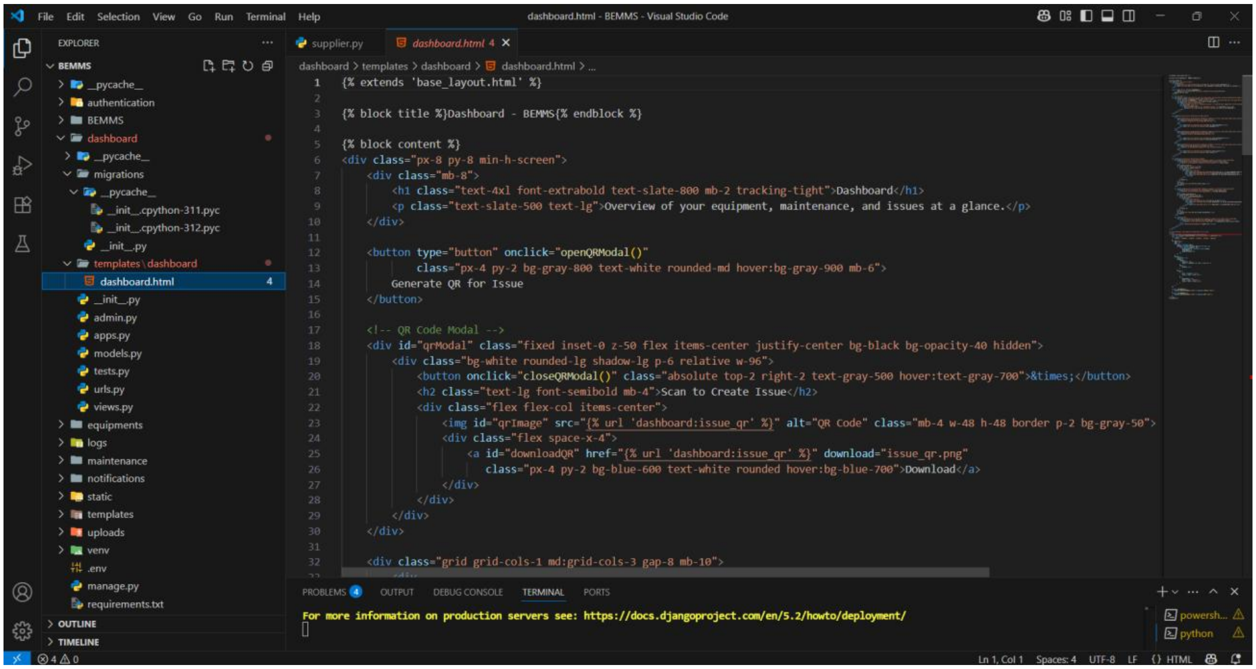
Python Software Foundation. (2025). Python Language Reference. https://www.python.org

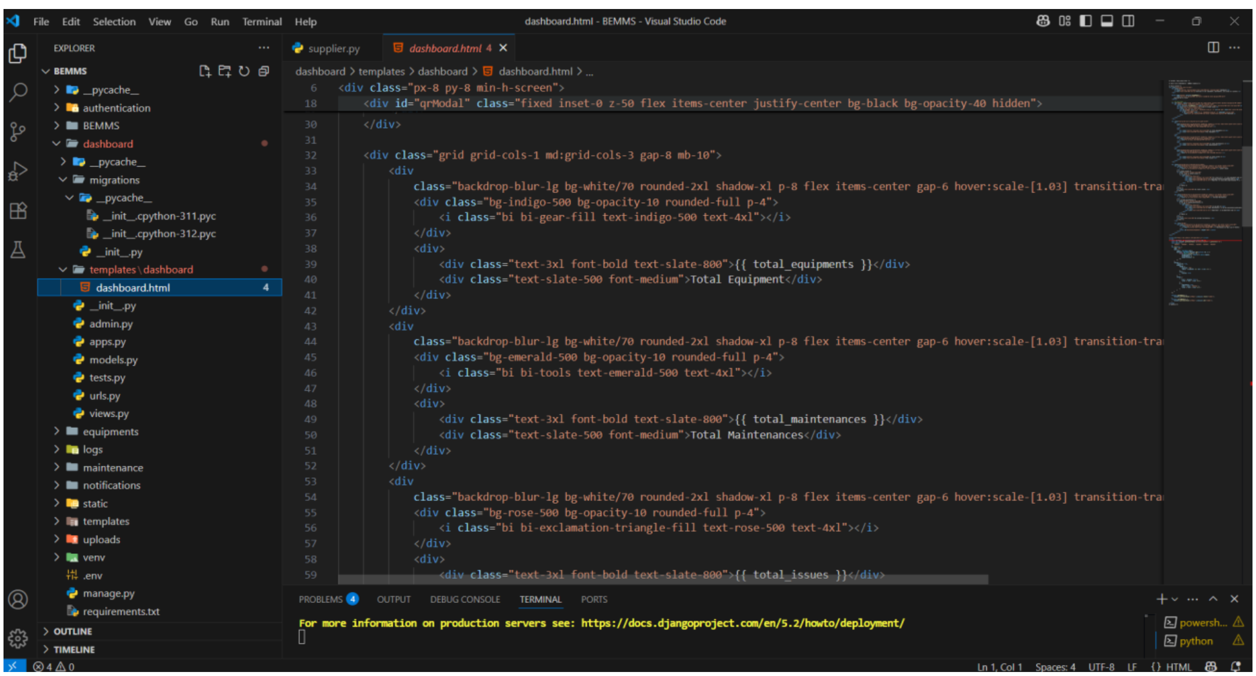
- Optional citation of Stack Overflow discussion (if allowed by your tutor):

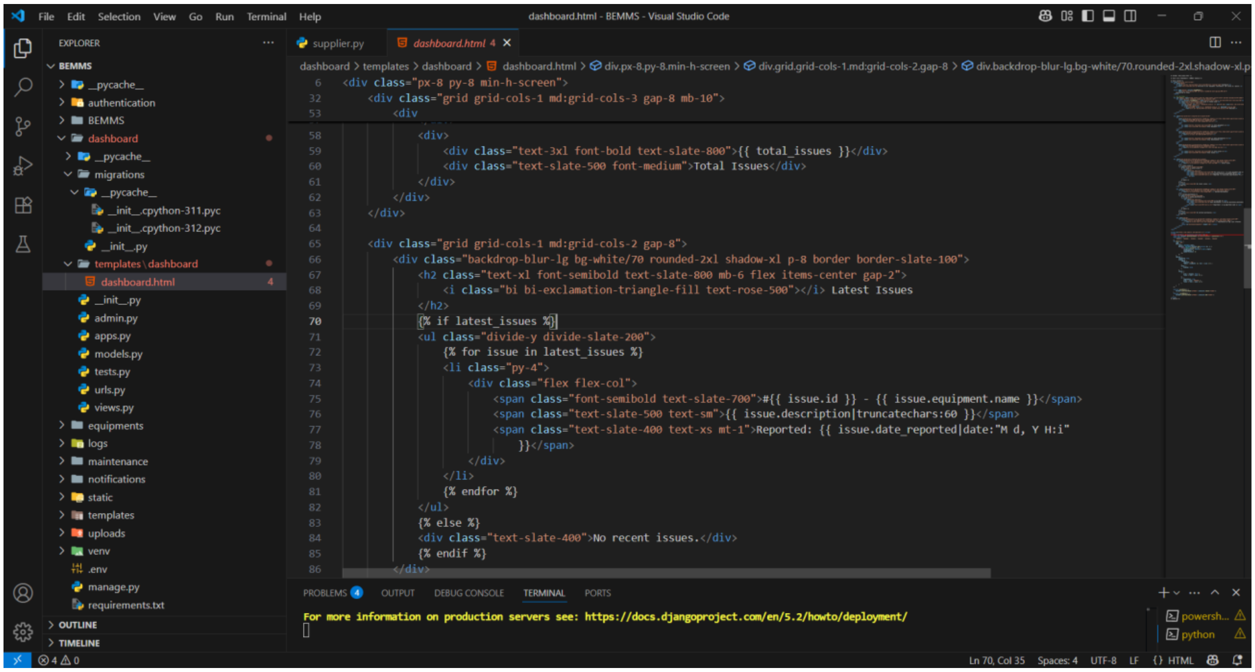
  Stack Overflow. (2025). Retrieved from https://stackoverflow.com

Annexures

Source Code



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dashboard.html 4
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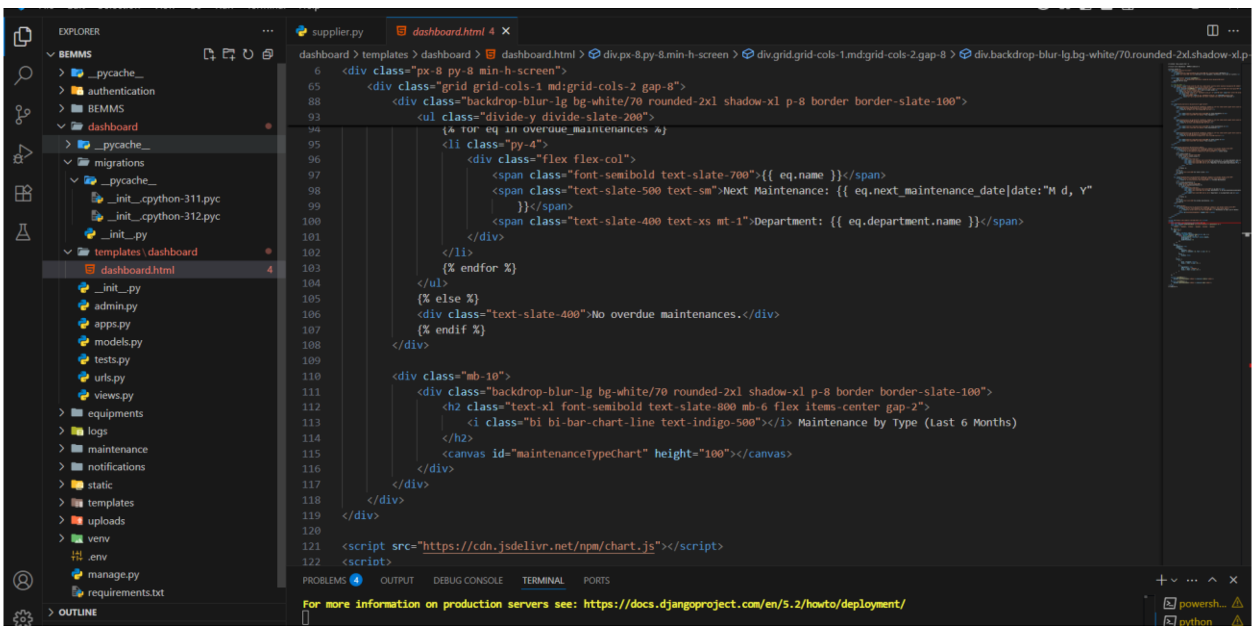
<div
<div>
<div class="text-3xl font-bold text-slate-800">{{ total_issues }}</div>
<div class="text-slate-500 font-medium">Total Issues</div>
</div>
</div>
</div>
<div class="grid grid-cols-1 md:grid-cols-2 gap-8">
<div class="backdrop-blur-lg bg-white/70 rounded-2xl shadow-xl p-8 border border-slate-100">
<h2 class="text-xl font-semibold text-slate-800 mb-6 flex items-center gap-2">
<i class="bi bi-exclamation-triangle-fill text-rose-500"></i> Latest Issues
</h2>
{% if latest_issues %}
<ul class="divide-y divide-slate-200">
{% for issue in latest_issues %}
<li class="py-4">
<div class="flex flex-col">
#{{ issue.id }} - {{ issue.equipment.name }}
{{ issue.description|truncatechars:60 }}
Reported: {{ issue.date_reported|date:"M d, Y H:i"
}}
</div>

{% endfor %}

{% else %}
<div class="text-slate-400">No recent issues.</div>
{% endif %}
</div>
PROBLEMS 4
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
For more information on production servers see: https://docs.djangoproject.com/en/5.2/howto/deployment/
Ln 70, Col 35
Spaces: 4
UTF-8
LF
{} HTML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<div class="grid grid-cols-1 md:grid-cols-2 gap-8">
<div class="backdrop-blur-lg bg-white/70 rounded-2xl shadow-xl p-8 border border-slate-100">
<ul class="divide-y divide-slate-200">
{% for eq in overdue_maintenances %}
<li class="py-4">
<div class="flex flex-col">
{{ eq.name }}
Next Maintenance: {{ eq.next_maintenance_date|date:"M d, Y"
}}
Department: {{ eq.department.name }}
</div>

{% endfor %}

{% else %}
<div class="text-slate-400">No overdue maintenances.</div>
{% endif %}
</div>
<div class="mb-10">
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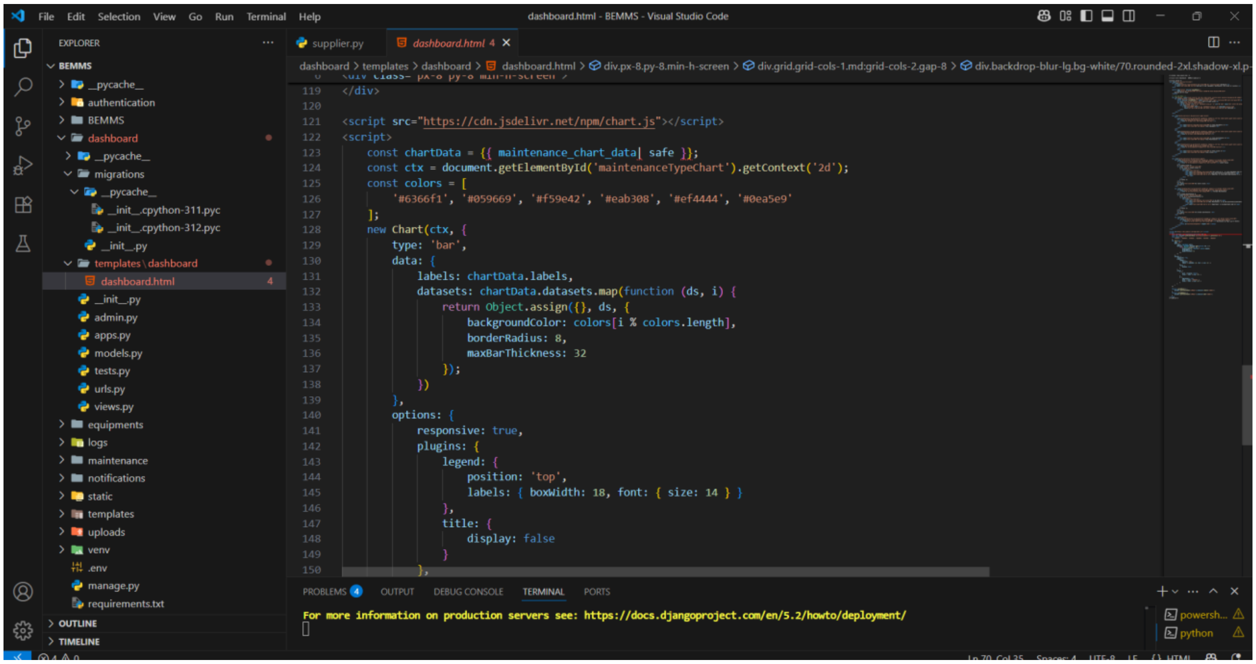
<h2 class="text-xl font-semibold text-slate-800 mb-6 flex items-center gap-2">
<i class="bi bi-bar-chart-line text-indigo-500"></i> Maintenance by Type (Last 6 Months)
</h2>
<canvas id="maintenanceTypeChart" height="100"></canvas>
</div>
</div>
</div>
</div>
<script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
<script>
PROBLEMS 4
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
For more information on production servers see: https://docs.djangoproject.com/en/5.2/howto/deployment/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</div>
<script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
<script>
const chartData = {{ maintenance_chart_data| safe }};
const ctx = document.getElementById('maintenanceTypeChart').getContext('2d');
const colors = [
'#6366f1', '#059669', '#f59e42', '#eab308', '#ef4444', '#0ea5e9'
];
new Chart(ctx, {
type: 'bar',
data: {
labels: chartData.labels,
datasets: chartData.datasets.map(function (ds, i) {
return Object.assign({}, ds, {
backgroundColor: colors[i % colors.length],
borderRadius: 8,
maxBarThickness: 32
});
})
},
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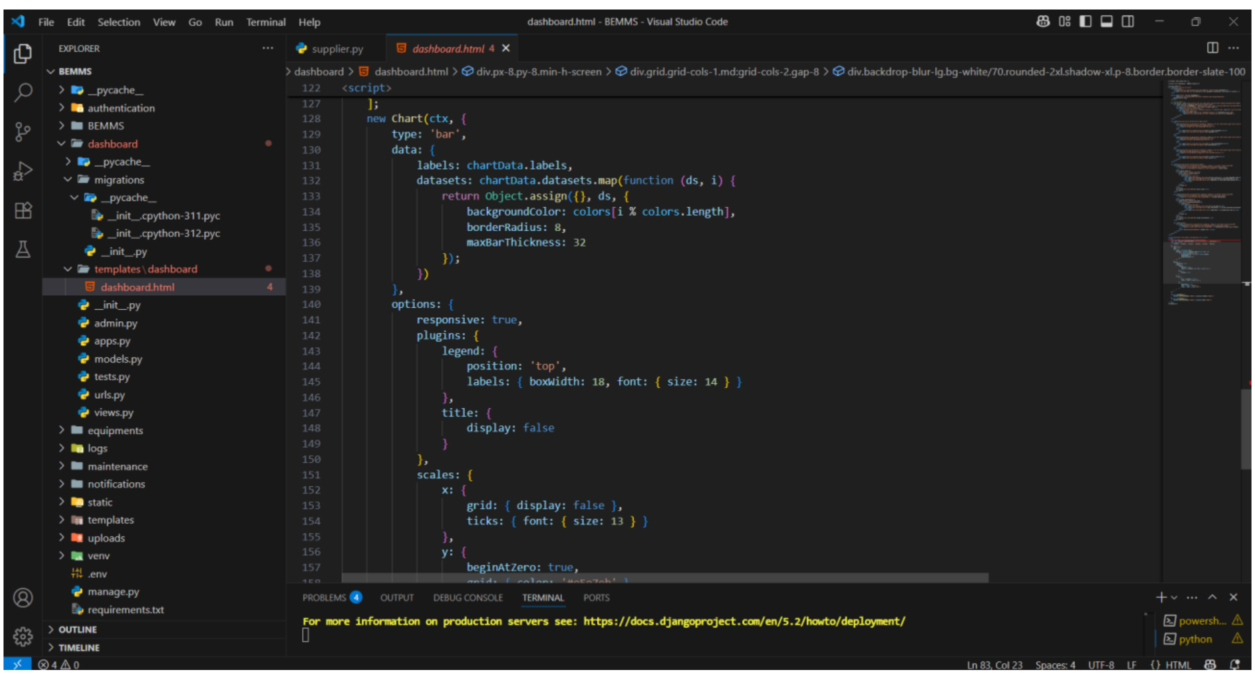
For more information on production servers see: https://docs.djangoproject.com/en/5.2/howto/deployment/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];
new Chart(ctx, {
type: 'bar',
data: {
labels: chartData.labels,
datasets: chartData.datasets.map(function (ds, i) {
return Object.assign({}, ds, {
backgroundColor: colors[i % colors.length],
borderRadius: 8,
maxBarThickness: 32
});
})
},
options: {
responsive: true,
plugins: {
legend: {
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labels: { boxWidth: 18, font: { size: 14 } }
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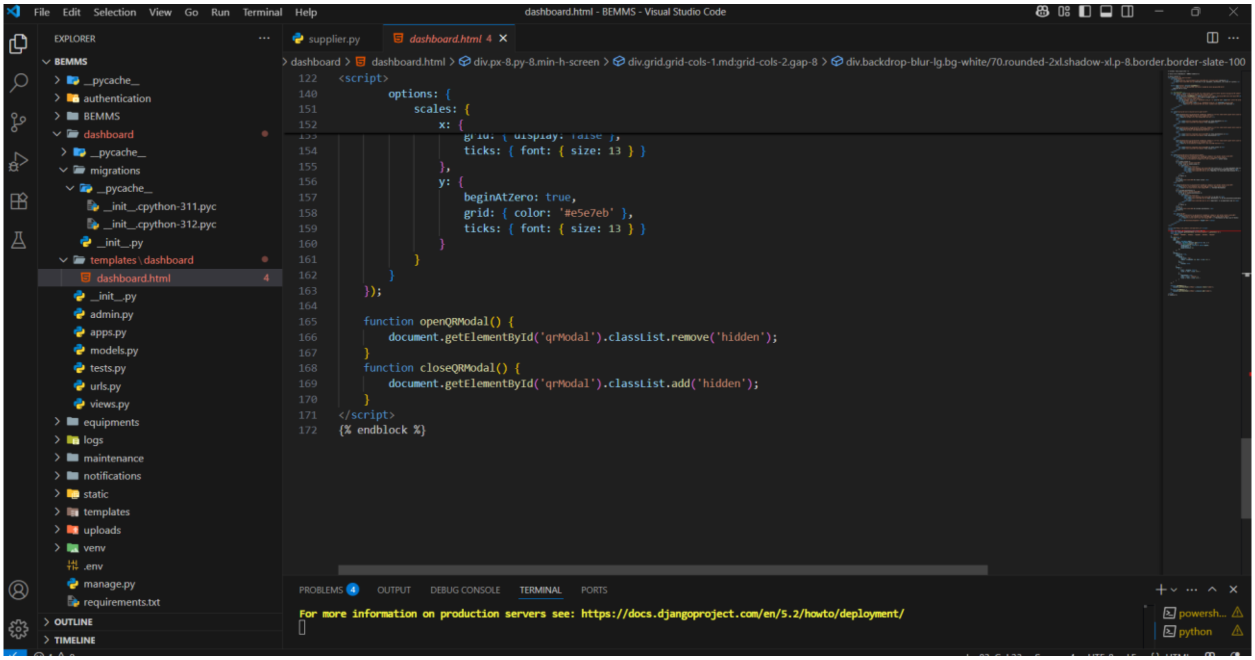
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dashboard.html 4
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x: {
ticks: { font: { size: 13 } }
},
y: {
beginAtZero: true,
grid: { color: '#e5e7eb' },
ticks: { font: { size: 13 } }
}
}
}
});
function openQRModal() {
document.getElementById('qrModal').classList.remove('hidden');
}
function closeQRModal() {
document.getElementById('qrModal').classList.add('hidden');
}
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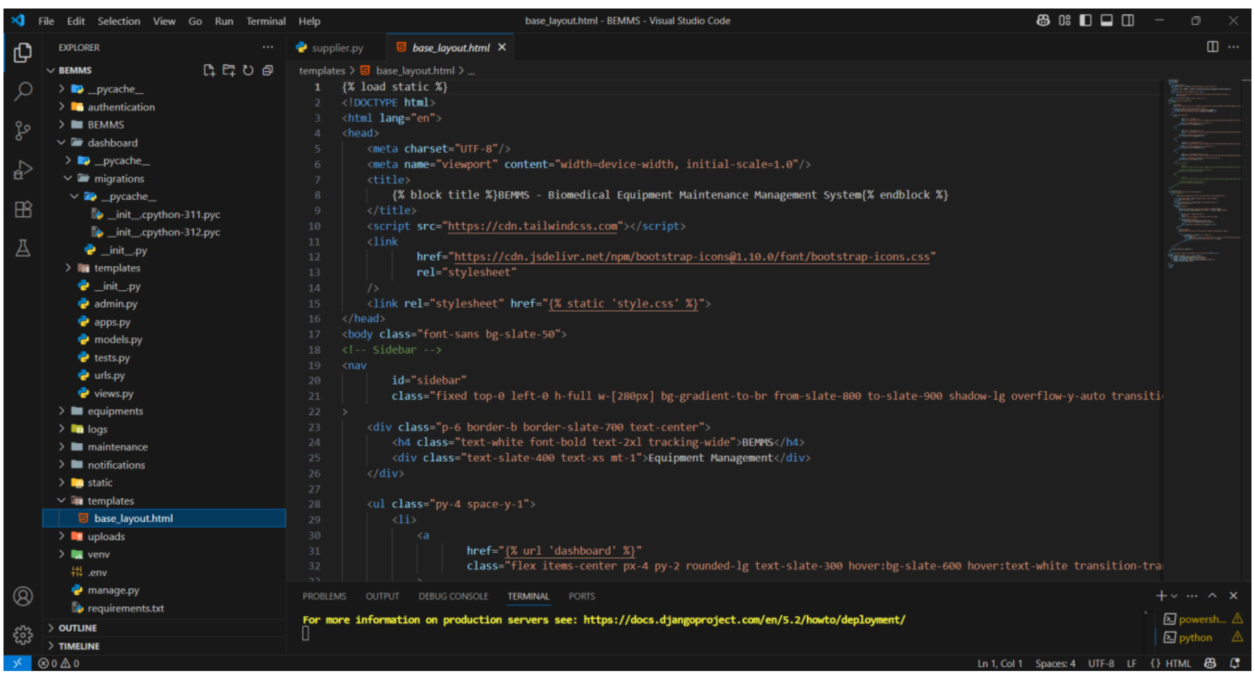
{% endblock %}
For more information on production servers see: https://docs.djangoproject.com/en/5.2/howto/deployment/

base layout


base_layout.html - BEMMS - Visual Studio Code
supplier.py
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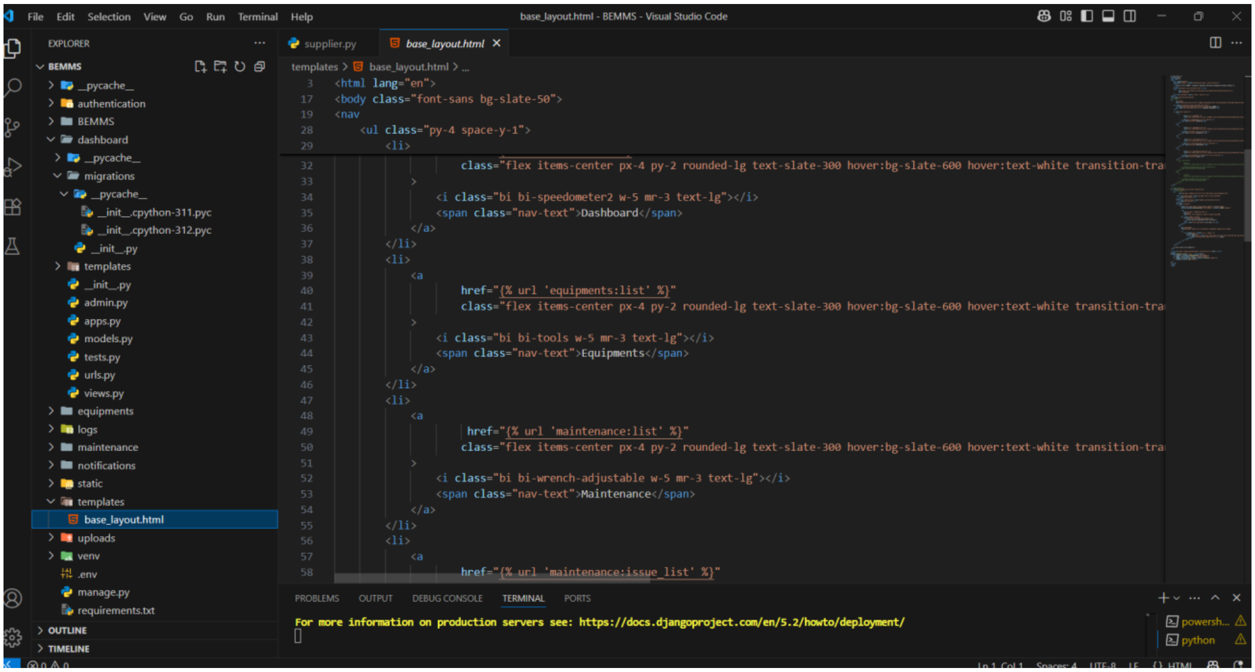
{% load static %}
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8"/>
<meta name="viewport" content="width=device-width, initial-scale=1.0"/>
<title>
{% block title %}BEMMS - Biomedical Equipment Maintenance Management System{% endblock %}
</title>
<script src="https://cdn.tailwindcss.com"></script>
<link
href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.10.0/font/bootstrap-icons.css"
rel="stylesheet"
/>
<link rel="stylesheet" href="{% static 'style.css' %}">
</head>
<body class="font-sans bg-slate-50">
<!-- Sidebar -->
<nav
id="sidebar"
class="fixed top-0 left-0 h-full w-[280px] bg-gradient-to-br from-slate-800 to-slate-900 shadow-lg overflow-y-auto transiti
>
<div class="p-6 border-b border-slate-700 text-center">
<h4 class="text-white font-bold text-2xl tracking-wide">BEMMS</h4>
<div class="text-slate-400 text-xs mt-1">Equipment Management</div>
</div>
<ul class="py-4 space-y-1">

<a
href="{% url 'dashboard' %}"
class="flex items-center px-4 py-2 rounded-lg text-slate-300 hover:bg-slate-600 hover:text-white transition-tra
For more information on production servers see: https://docs.djangoproject.com/en/5.2/howto/deployment/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supplier.py
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templates > base_layout.html > ...
<html lang="en">
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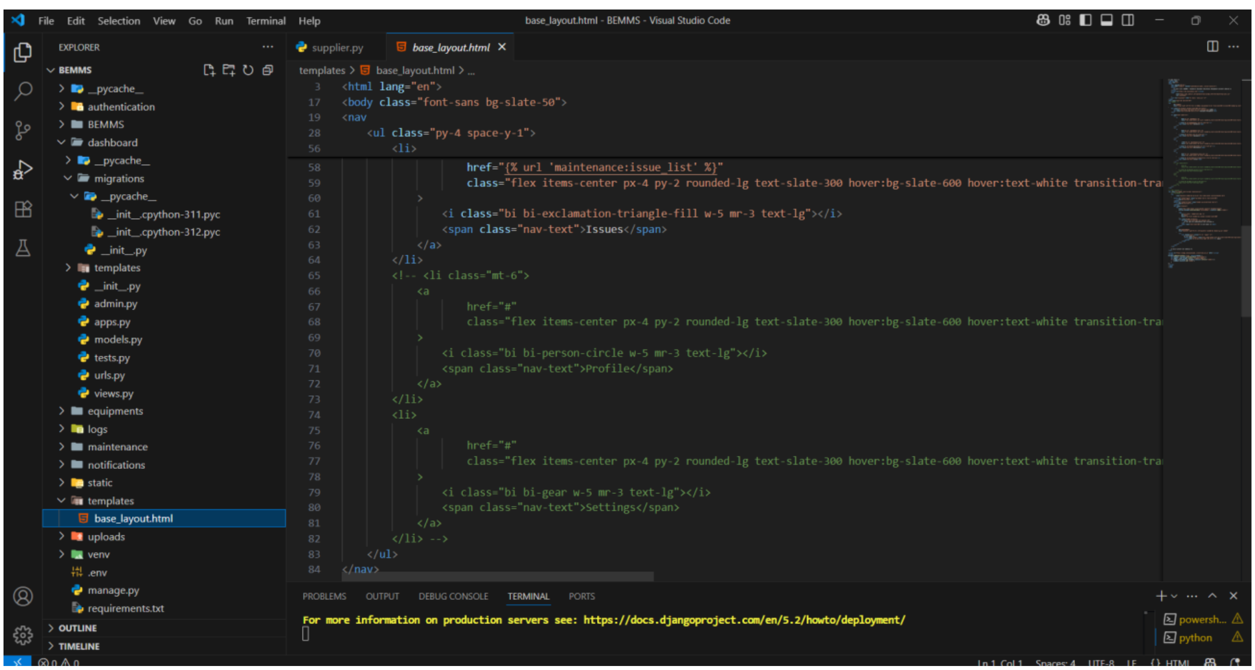
<nav
<ul class="py-4 space-y-1">

class="flex items-center px-4 py-2 rounded-lg text-slate-300 hover:bg-slate-600 hover:text-white transition-tra
>
<i class="bi bi-speedometer2 w-5 mr-3 text-lg"></i>
Dashboard

<a
href="{% url 'equipments:list' %}"
class="flex items-center px-4 py-2 rounded-lg text-slate-300 hover:bg-slate-600 hover:text-white transition-tra
>
<i class="bi bi-tools w-5 mr-3 text-lg"></i>
Equipments

<a
href="{% url 'maintenance:list' %}"
class="flex items-center px-4 py-2 rounded-lg text-slate-300 hover:bg-slate-600 hover:text-white transition-tra
>
<i class="bi bi-wrench-adjustable w-5 mr-3 text-lg"></i>
Maintenance

<a
href="{% url 'maintenance:issue_list' %}"
For more information on production servers see: https://docs.djangoproject.com/en/5.2/howto/deployment/


base_layout.html - BEMMS - Visual Studio Code
supplier.py
base_layout.html
templates > base_layout.html > ...
<html lang="en">
<body class="font-sans bg-slate-50">
<nav
<ul class="py-4 space-y-1">

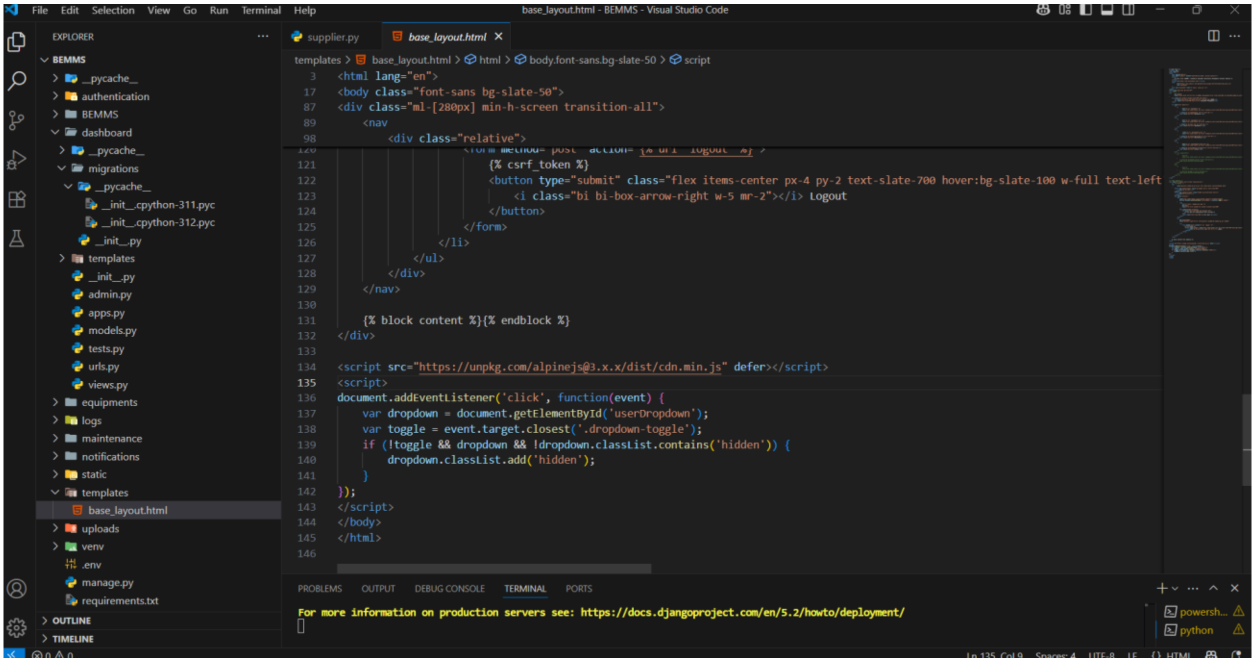
href="{% url 'maintenance:issue_list' %}"
class="flex items-center px-4 py-2 rounded-lg text-slate-300 hover:bg-slate-600 hover:text-white transition-tra
>
<i class="bi bi-exclamation-triangle-fill w-5 mr-3 text-lg"></i>
Issues

<!-- <li class="mt-6">
<a
href="#"
class="flex items-center px-4 py-2 rounded-lg text-slate-300 hover:bg-slate-600 hover:text-white transition-tra
>
<i class="bi bi-person-circle w-5 mr-3 text-lg"></i>
Profile

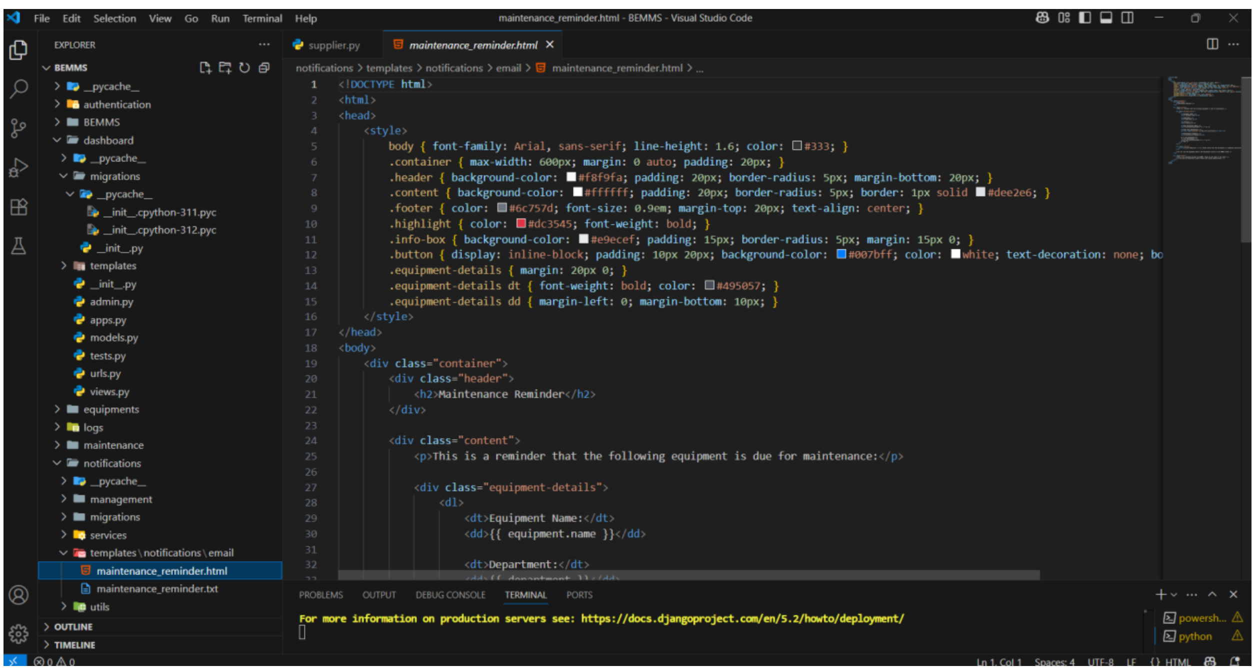
<a
href="#"
class="flex items-center px-4 py-2 rounded-lg text-slate-300 hover:bg-slate-600 hover:text-white transition-tra
>
<i class="bi bi-gear w-5 mr-3 text-lg"></i>
Settings

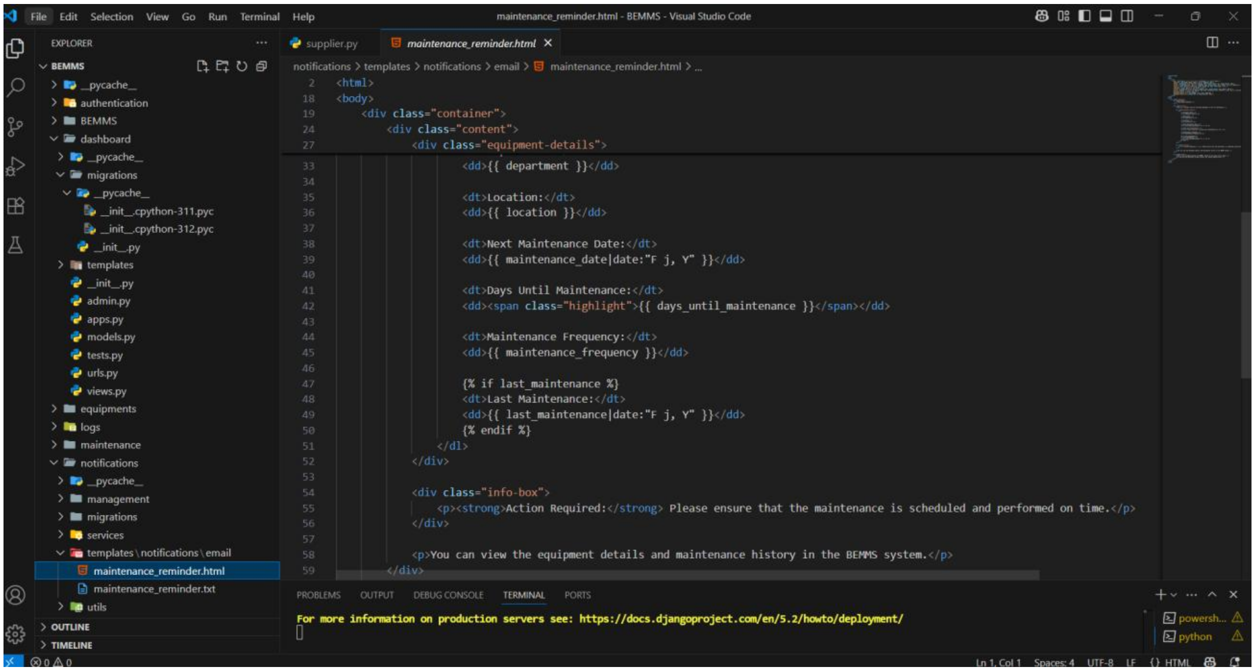
 -->

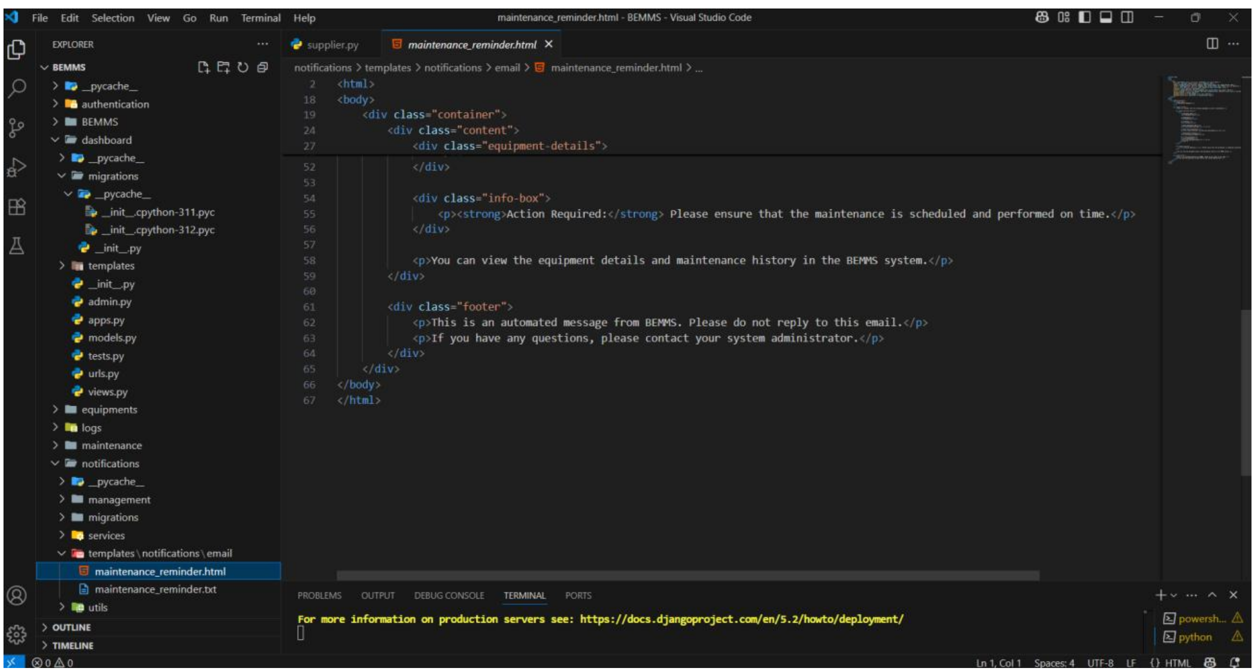
</nav>
For more information on production servers see: https://docs.djangoproject.com/en/5.2/howto/deployment/



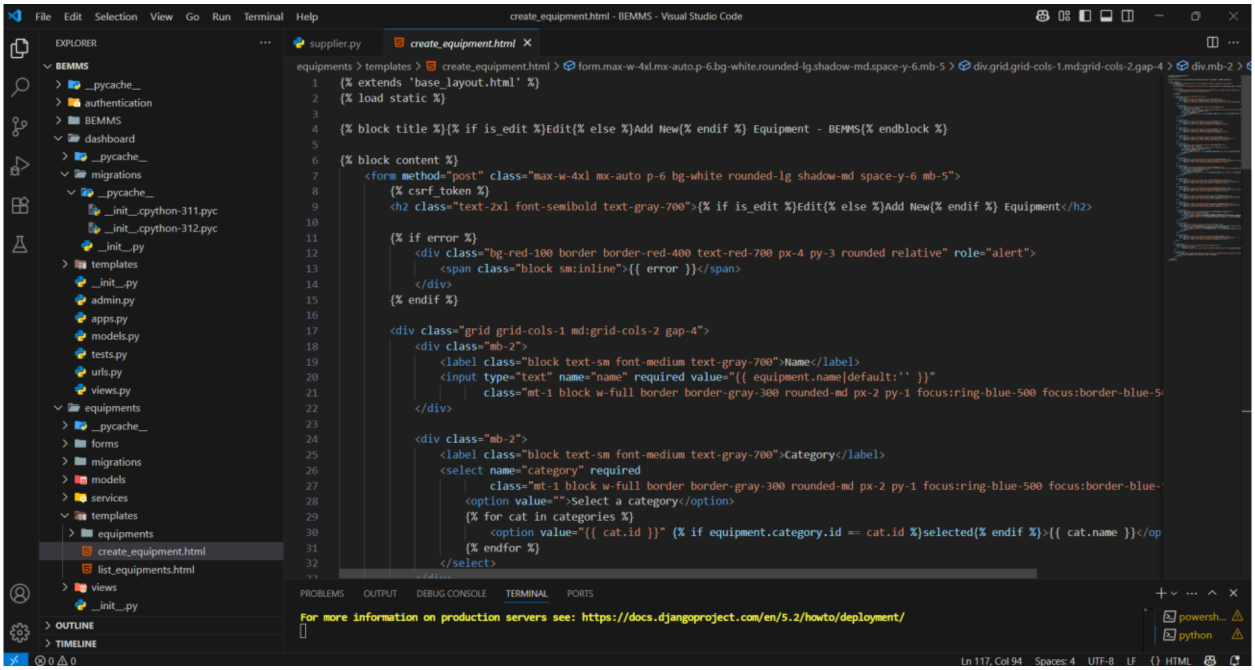
Maintenance Reminder




maintenance_reminder.html - BEMMS - Visual Studio Code
supplier.py
maintenance_reminder.html
notifications > templates > notifications > email > maintenance_reminder.html > ...
<html>
<body>
<div class="container">
<div class="content">
<div class="equipment-details">
<dd>{{ department }}</dd>
<dt>Location:</dt>
<dd>{{ location }}</dd>
<dt>Next Maintenance Date:</dt>
<dd>{{ maintenance_date|date:"F j, Y" }}</dd>
<dt>Days Until Maintenance:</dt>
<dd>{{ days_until_maintenance }}</dd>
<dt>Maintenance Frequency:</dt>
<dd>{{ maintenance_frequency }}</dd>
{% if last_maintenance %}
<dt>Last Maintenance:</dt>
<dd>{{ last_maintenance|date:"F j, Y" }}</dd>
{% endif %}
</dl>
</div>
<div class="info-box">
<p>Action Required: Please ensure that the maintenance is scheduled and performed on time.</p>
</div>
<p>You can view the equipment details and maintenance history in the BEMMS system.</p>
</div>
For more information on production servers see: https://docs.djangoproject.com/en/5.2/howto/deployment/


maintenance_reminder.html - BEMMS - Visual Studio Code
notifications > templates > notifications > email > maintenance_reminder.html > ...
<html>
<body>
<div class="container">
<div class="content">
<div class="equipment-details">
</div>
<div class="info-box">
<p>Action Required: Please ensure that the maintenance is scheduled and performed on time.</p>
</div>
<p>You can view the equipment details and maintenance history in the BEMMS system.</p>
</div>
<div class="footer">
<p>This is an automated message from BEMMS. Please do not reply to this email.</p>
<p>If you have any questions, please contact your system administrator.</p>
</div>
</div>
</body>
</html>
For more information on production servers see: https://docs.djangoproject.com/en/5.2/howto/deployment/

Equipment


create_equipment.html - BEMMS - Visual Studio Code
supplier.py
create_equipment.html
{% extends 'base_layout.html' %}
{% load static %}
{% block title %}{% if is_edit %}Edit{% else %}Add New{% endif %} Equipment - BEMMS{% endblock %}
{% block content %}
<form method="post" class="max-w-4xl mx-auto p-6 bg-white rounded-lg shadow-md space-y-6 mb-5">
{% csrf_token %}
<h2 class="text-2xl font-semibold text-gray-700">{% if is_edit %}Edit{% else %}Add New{% endif %} Equipment</h2>
{% if error %}
<div class="bg-red-100 border border-red-400 text-red-700 px-4 py-3 rounded relative" role="alert">
{{ error }}
</div>
{% endif %}
<div class="grid grid-cols-1 md:grid-cols-2 gap-4">
<div class="mb-2">
<label class="block text-sm font-medium text-gray-700">Name</label>
<input type="text" name="name" required value="{{ equipment.name|default:'' }}"
</div>
<div class="mb-2">
<label class="block text-sm font-medium text-gray-700">Category</label>
<select name="category" required
<option value="">Select a category</option>
{% for cat in categories %}
{% endfor %}
</select>
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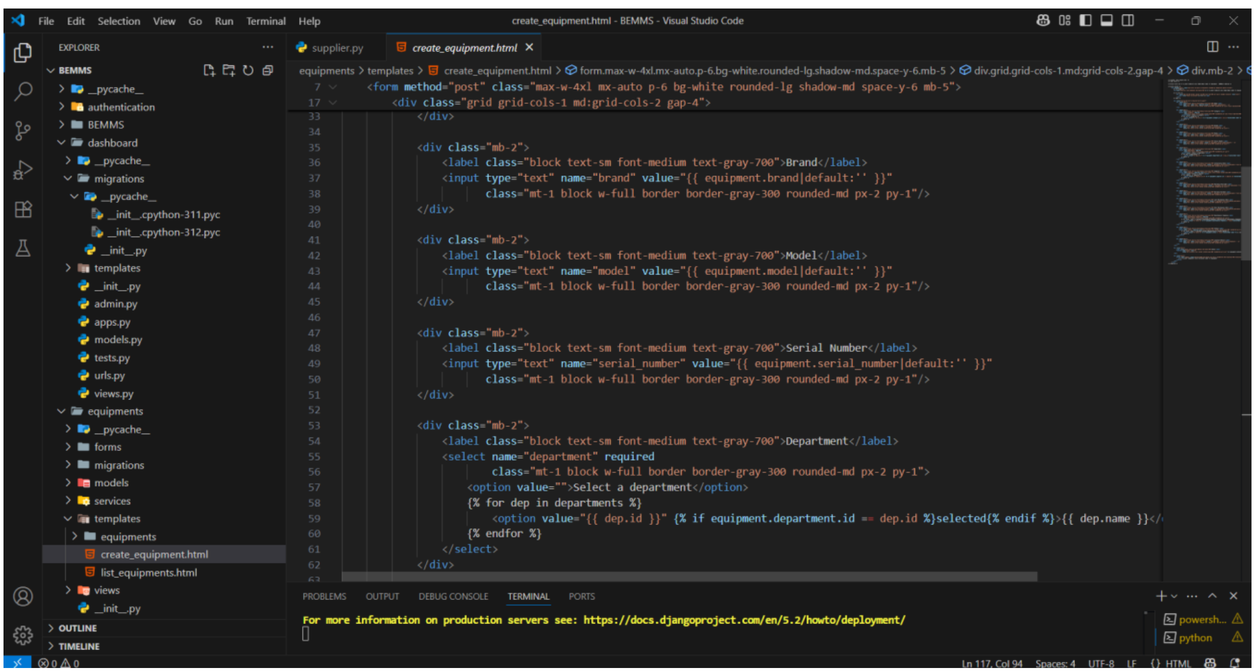
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<div class="mb-2">
<label class="block text-sm font-medium text-gray-700">Brand</label>
<input type="text" name="brand" value="{{ equipment.brand|default:'' }}"
class="mt-1 block w-full border border-gray-300 rounded-md px-2 py-1"/>
</div>
<div class="mb-2">
<label class="block text-sm font-medium text-gray-700">Model</label>
<input type="text" name="model" value="{{ equipment.model|default:'' }}"
class="mt-1 block w-full border border-gray-300 rounded-md px-2 py-1"/>
</div>
<div class="mb-2">
<label class="block text-sm font-medium text-gray-700">Serial Number</label>
<input type="text" name="serial_number" value="{{ equipment.serial_number|default:'' }}"
class="mt-1 block w-full border border-gray-300 rounded-md px-2 py-1"/>
</div>
<div class="mb-2">
<label class="block text-sm font-medium text-gray-700">Department</label>
<select name="department" required
class="mt-1 block w-full border border-gray-300 rounded-md px-2 py-1">
<option value="">Select a department</option>
{% for dep in departments %}
{% endfor %}
</select>
</div>
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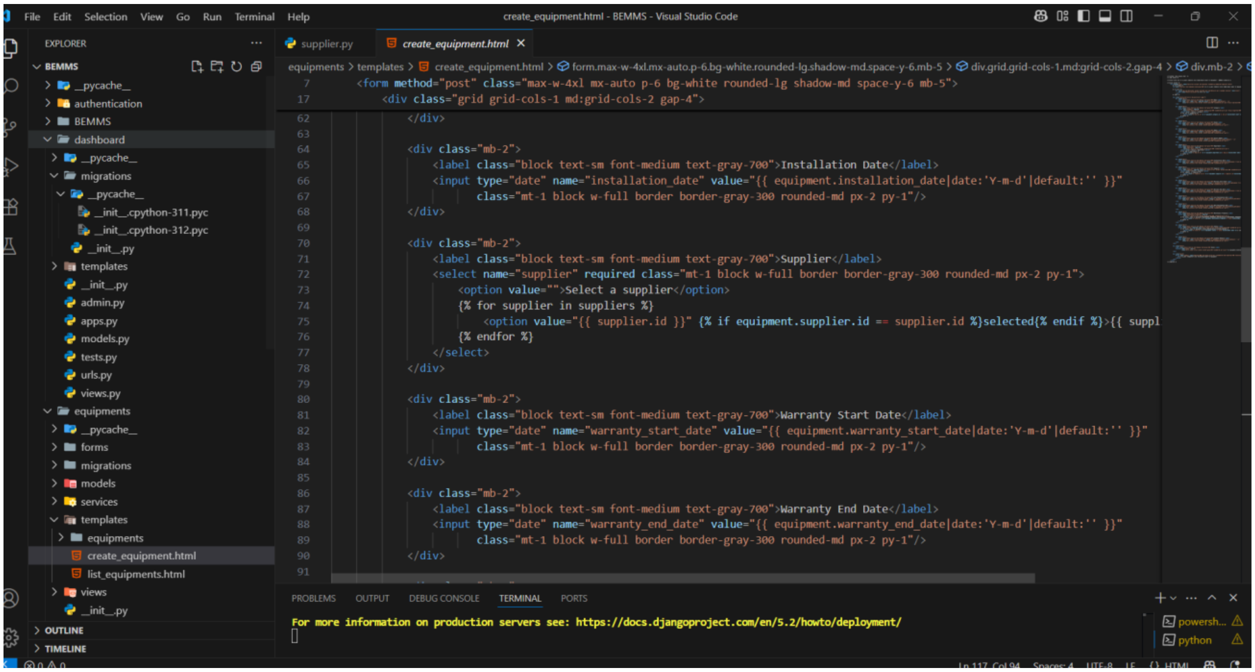
create_equipment.html - BEMMS - Visual Studio Code
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equipments > templates > create_equipment.html > form.max-w-4xl.mx-auto.p-6.bg-white.rounded-lg.shadow-md.space-y-6.mb-5 > div.grid.grid-cols-1.md:grid-cols-2.gap-4 > div.mb-2
<form method="post" class="max-w-4xl mx-auto p-6 bg-white rounded-lg shadow-md space-y-6 mb-5">
<div class="grid grid-cols-1 md:grid-cols-2 gap-4">
</div>
<div class="mb-2">
<label class="block text-sm font-medium text-gray-700">Installation Date</label>
<input type="date" name="installation_date" value="{{ equipment.installation_date|date:'Y-m-d'|default:'' }}"
class="mt-1 block w-full border border-gray-300 rounded-md px-2 py-1"/>
</div>
<div class="mb-2">
<label class="block text-sm font-medium text-gray-700">Supplier</label>
<select name="supplier" required class="mt-1 block w-full border border-gray-300 rounded-md px-2 py-1">
<option value="">Select a supplier</option>
{% for supplier in suppliers %}
<option value="{{ supplier.id }}" {% if equipment.supplier.id == supplier.id %}selected{% endif %}>{{ suppl
{% endfor %}
</select>
</div>
<div class="mb-2">
<label class="block text-sm font-medium text-gray-700">Warranty Start Date</label>
<input type="date" name="warranty_start_date" value="{{ equipment.warranty_start_date|date:'Y-m-d'|default:'' }}"
class="mt-1 block w-full border border-gray-300 rounded-md px-2 py-1"/>
</div>
<div class="mb-2">
<label class="block text-sm font-medium text-gray-700">Warranty End Date</label>
<input type="date" name="warranty_end_date" value="{{ equipment.warranty_end_date|date:'Y-m-d'|default:'' }}"
class="mt-1 block w-full border border-gray-300 rounded-md px-2 py-1"/>
</div>
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
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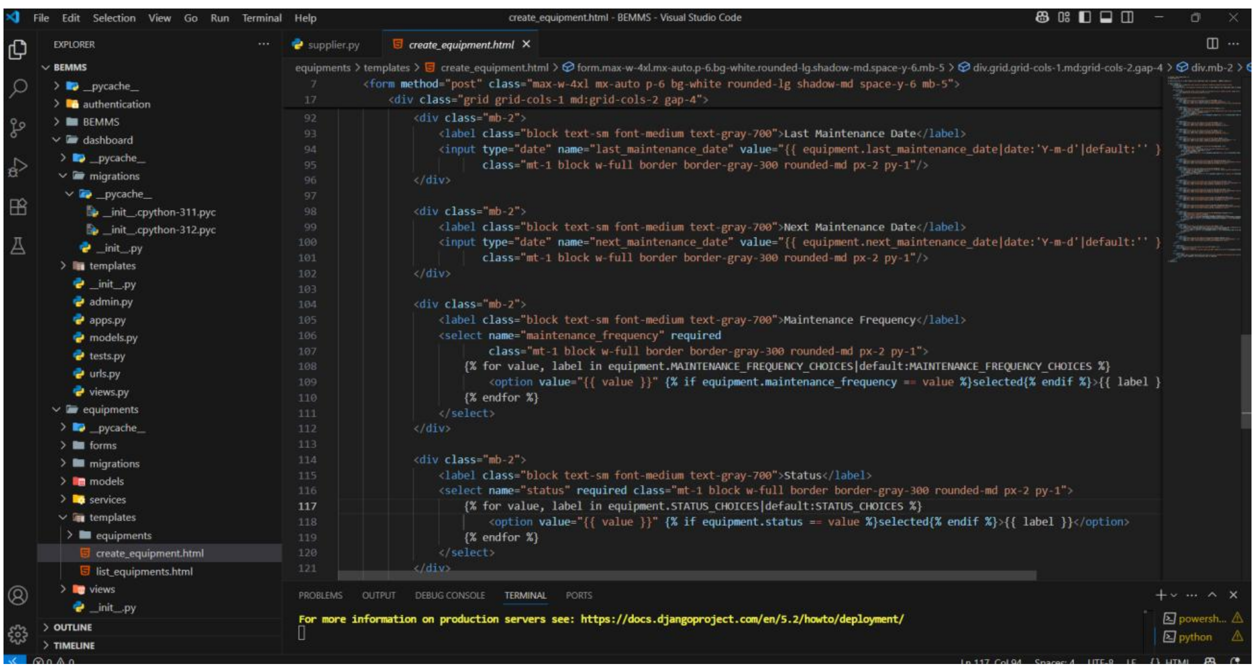
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<form method="post" class="max-w-4xl mx-auto p-6 bg-white rounded-lg shadow-md space-y-6 mb-5">
<div class="grid grid-cols-1 md:grid-cols-2 gap-4">
<div class="mb-2">
<label class="block text-sm font-medium text-gray-700">Last Maintenance Date</label>
<input type="date" name="last_maintenance_date" value="{{ equipment.last_maintenance_date|date:'Y-m-d'|default:'' }
class="mt-1 block w-full border border-gray-300 rounded-md px-2 py-1"/>
</div>
<div class="mb-2">
<label class="block text-sm font-medium text-gray-700">Next Maintenance Date</label>
<input type="date" name="next_maintenance_date" value="{{ equipment.next_maintenance_date|date:'Y-m-d'|default:'' }
class="mt-1 block w-full border border-gray-300 rounded-md px-2 py-1"/>
</div>
<div class="mb-2">
<label class="block text-sm font-medium text-gray-700">Maintenance Frequency</label>
<select name="maintenance_frequency" required
class="mt-1 block w-full border border-gray-300 rounded-md px-2 py-1">
{% for value, label in equipment.MAINTENANCE_FREQUENCY_CHOICES|default:MAINTENANCE_FREQUENCY_CHOICES %}
<option value="{{ value }}" {% if equipment.maintenance_frequency == value %}selected{% endif %}>{{ label }
{% endfor %}
</select>
</div>
<div class="mb-2">
<label class="block text-sm font-medium text-gray-700">Status</label>
<select name="status" required class="mt-1 block w-full border border-gray-300 rounded-md px-2 py-1">
{% for value, label in equipment.STATUS_CHOICES|default:STATUS_CHOICES %}
<option value="{{ value }}" {% if equipment.status == value %}selected{% endif %}>{{ label }}</option>
{% endfor %}
</select>
</div>
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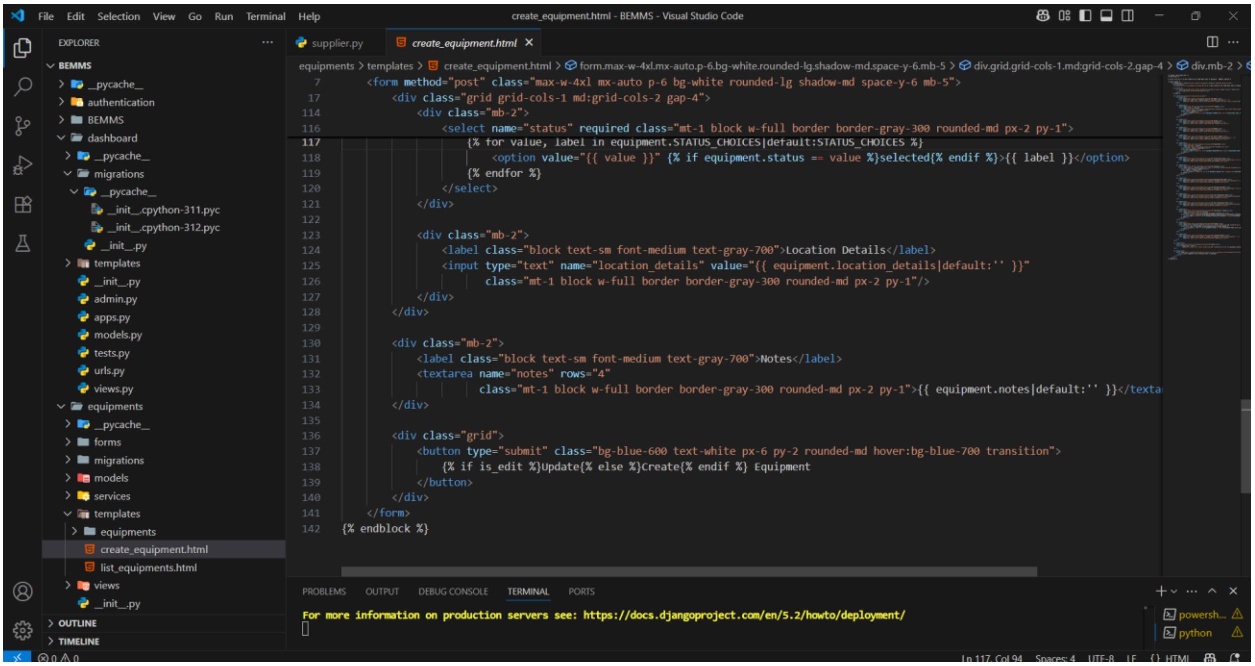
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<form method="post" class="max-w-4xl mx-auto p-6 bg-white rounded-lg shadow-md space-y-6 mb-5">
<div class="grid grid-cols-1 md:grid-cols-2 gap-4">
<div class="mb-2">
<select name="status" required class="mt-1 block w-full border border-gray-300 rounded-md px-2 py-1">
{% for value, label in equipment.STATUS_CHOICES|default:STATUS_CHOICES %}
<option value="{{ value }}" {% if equipment.status == value %}selected{% endif %}>{{ label }}</option>
{% endfor %}
</select>
</div>
<div class="mb-2">
<label class="block text-sm font-medium text-gray-700">Location Details</label>
<input type="text" name="location_details" value="{{ equipment.location_details|default:'' }}"
class="mt-1 block w-full border border-gray-300 rounded-md px-2 py-1"/>
</div>
</div>
<div class="mb-2">
<label class="block text-sm font-medium text-gray-700">Notes</label>
<textarea name="notes" rows="4"
class="mt-1 block w-full border border-gray-300 rounded-md px-2 py-1">{{ equipment.notes|default:'' }}</texta
</div>
<div class="grid">
<button type="submit" class="bg-blue-600 text-white px-6 py-2 rounded-md hover:bg-blue-700 transition">
{% if is_edit %}Update{% else %}Create{% endif %} Equipment
</button>
</div>
</form>
{% endblock %}
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